Improving Cost-Benefit Trade-off of LLM-based Agents under Self-Consistency with Early Termination and Model Hotswap

자기 일관성 기반 거대 언어 모델의 조기 종료 및 동적 교체를 통한 비용 대비 성능 개선

Naryeong Kim (Supervisor: Shin Yoo)

2026.07.14. ERC 여름 워크샵

LLM-based Agent Models

Large Language Models

LLM-based Agent Models

- Reasoning and solving complex problems including software engineering
- Prompt engineering techniques

Prompt Engineering

Problem

Alice has 3 apples. She buys 2 more packs of apples, each containing 4 apples.
Then she gives 5 apples to Bob. How many apples does she have left?

Prompt Engineering

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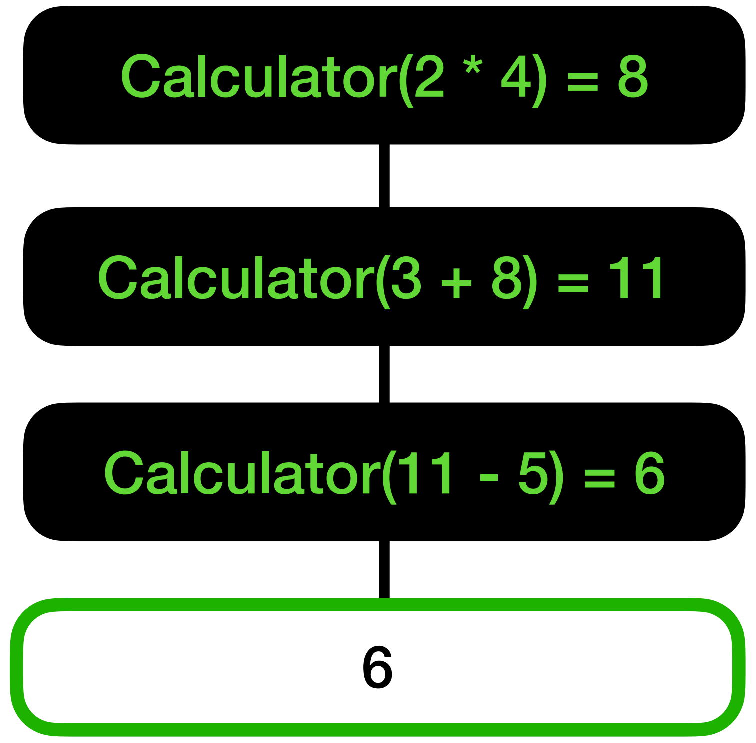
ReAct



Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, et al. 2023. ReAct: Synergizing Reasoning and Acting in Language Models. In Proceedings of the International Conference on Learning Representation (ICLR 2023).

Prompt Engineering

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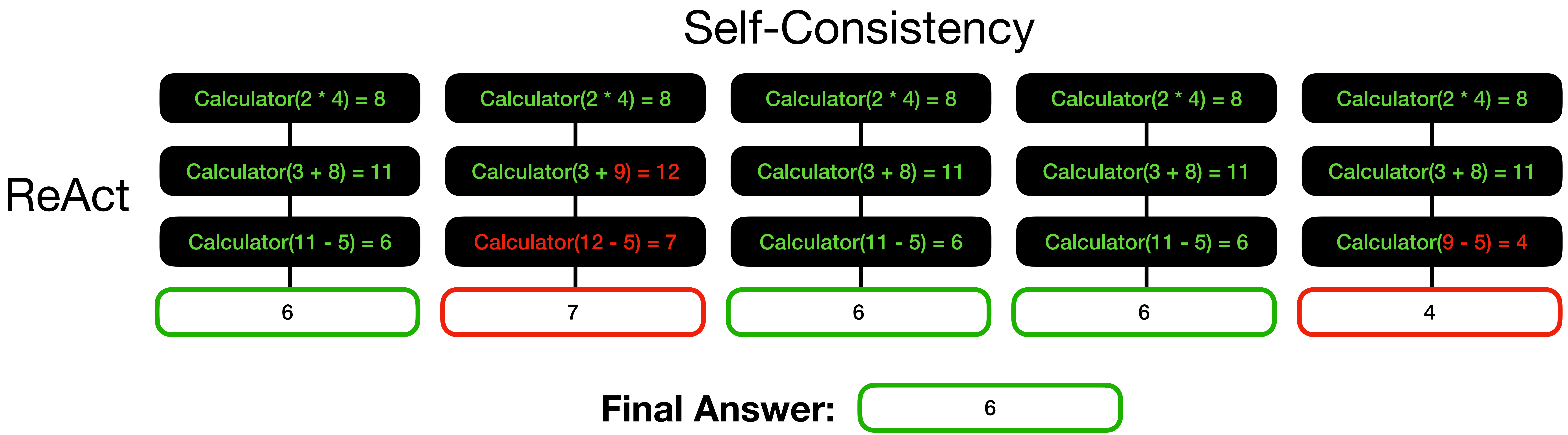

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, et al. 2023a. Self-Consistency Improves Chain of Thought Reasoning in Language Models. CoRR abs/2203.11171 (2023).

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Prompt Engineering

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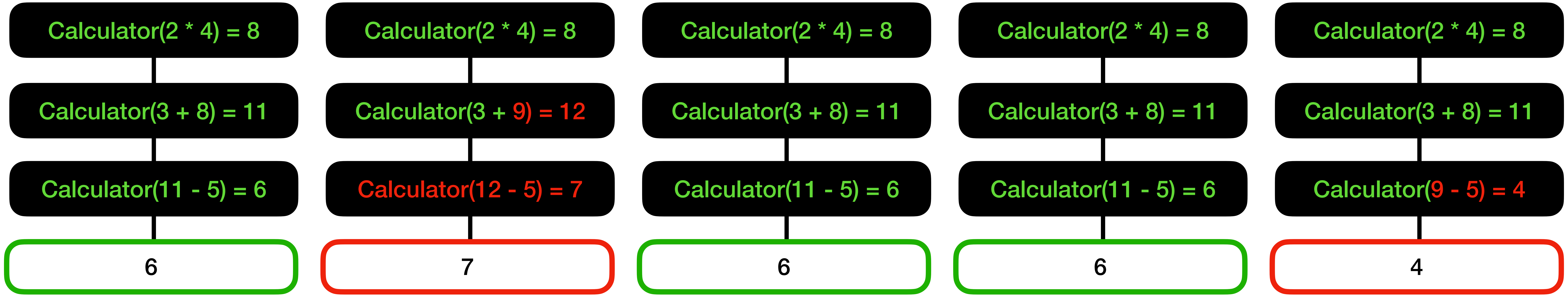
Self-Consistency

ReAct



Final Answer: 6

Performance gains from iterative reasoning step come with a cost penalty

Agent Base Model

Proprietary Models (LLMs)

Claude
Gemini
ChatGPT

- Large and Powerful
- High financial and computational cost

VS

Small Language Models (SLMs)

MISTRAL AI_
Meta AI
deepseek

- Less powerful
- Low latency and cost

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The cost-benefit trade-off agentic systems remains largely unexplored

Atropos (ISSTA 2026)

A predictive early-termination and LLM hot swapping technique running on a local SLM

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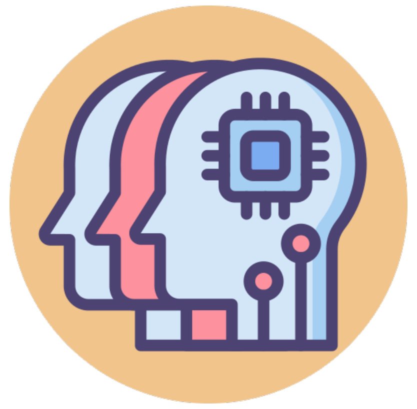

Agent

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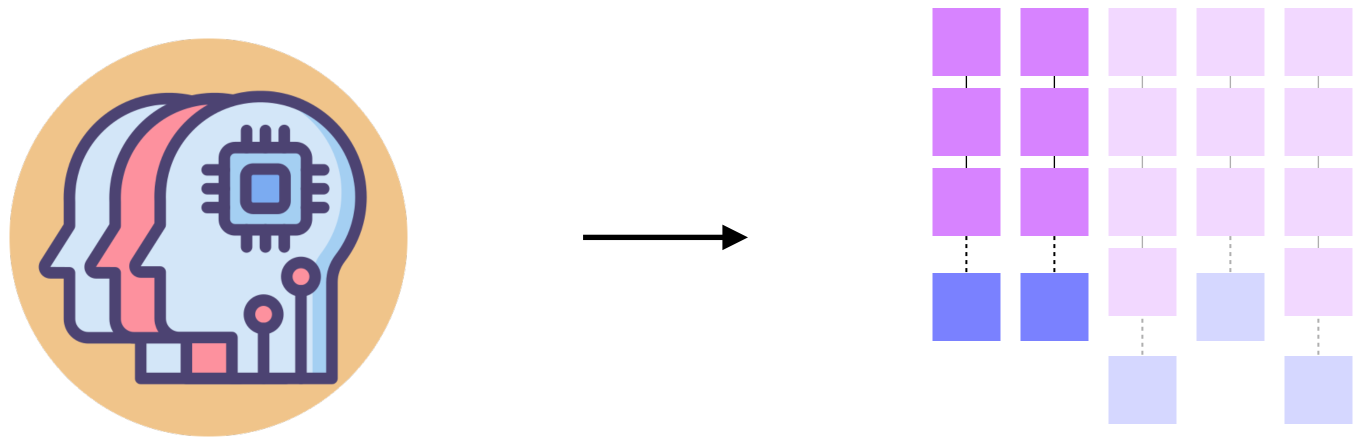

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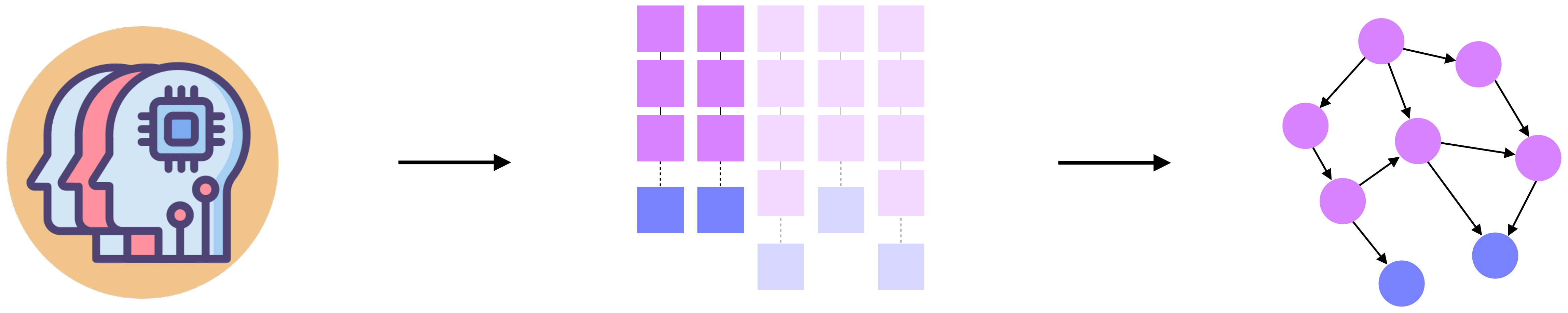

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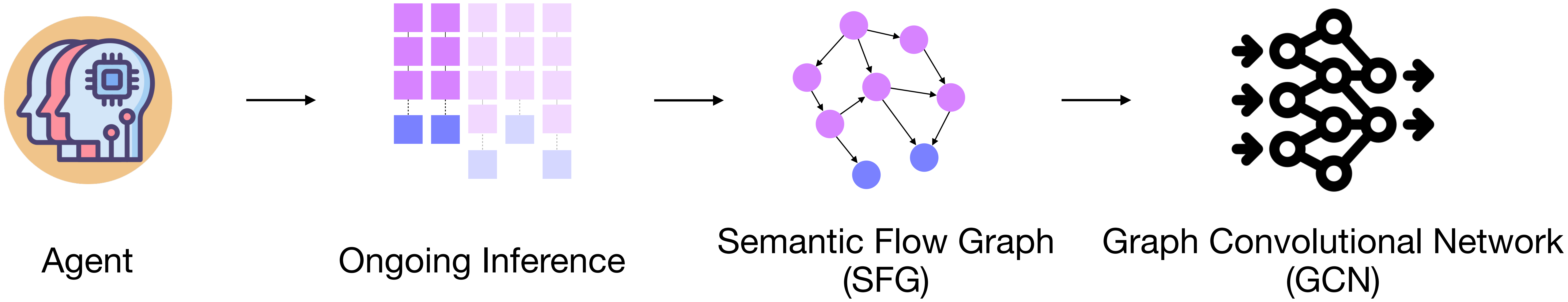

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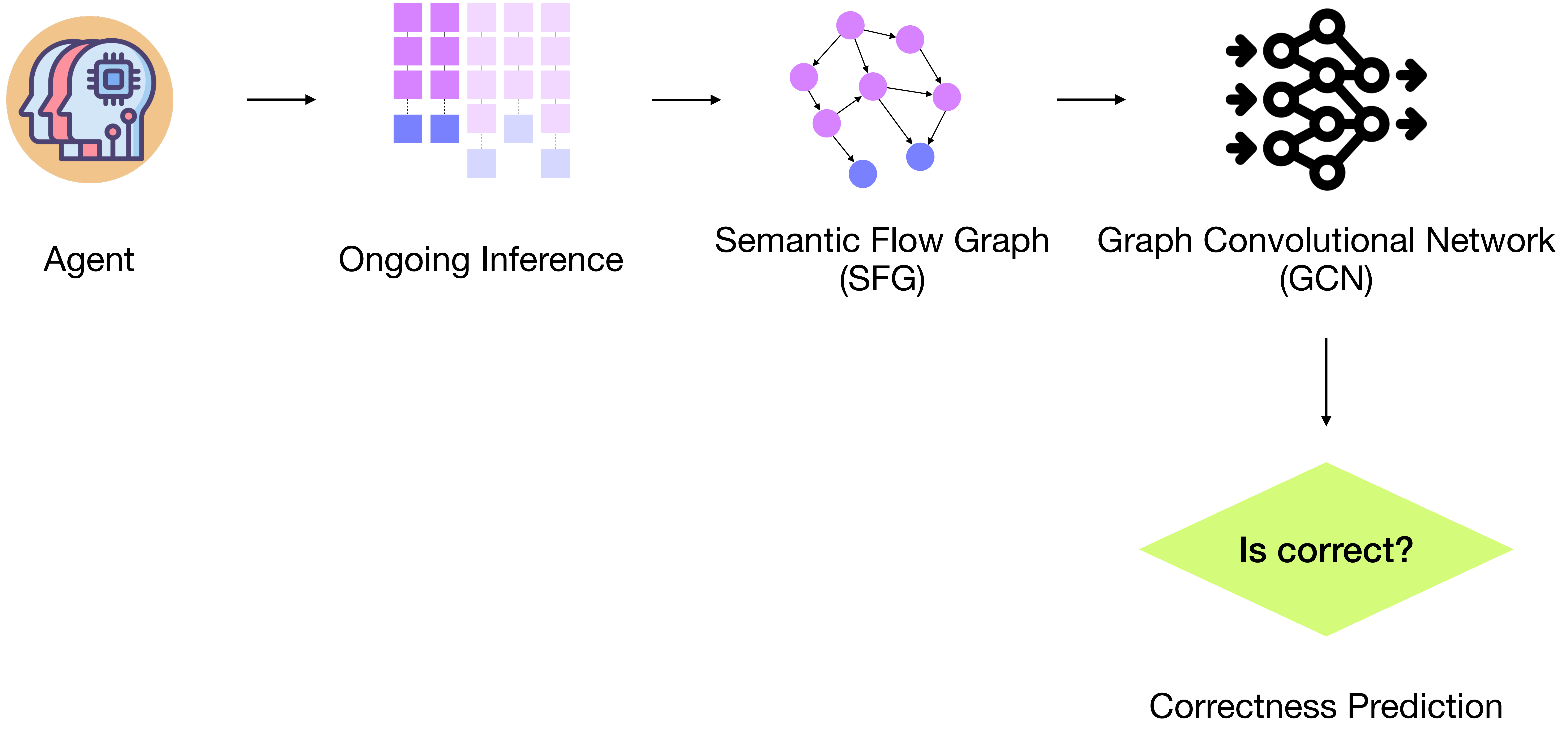

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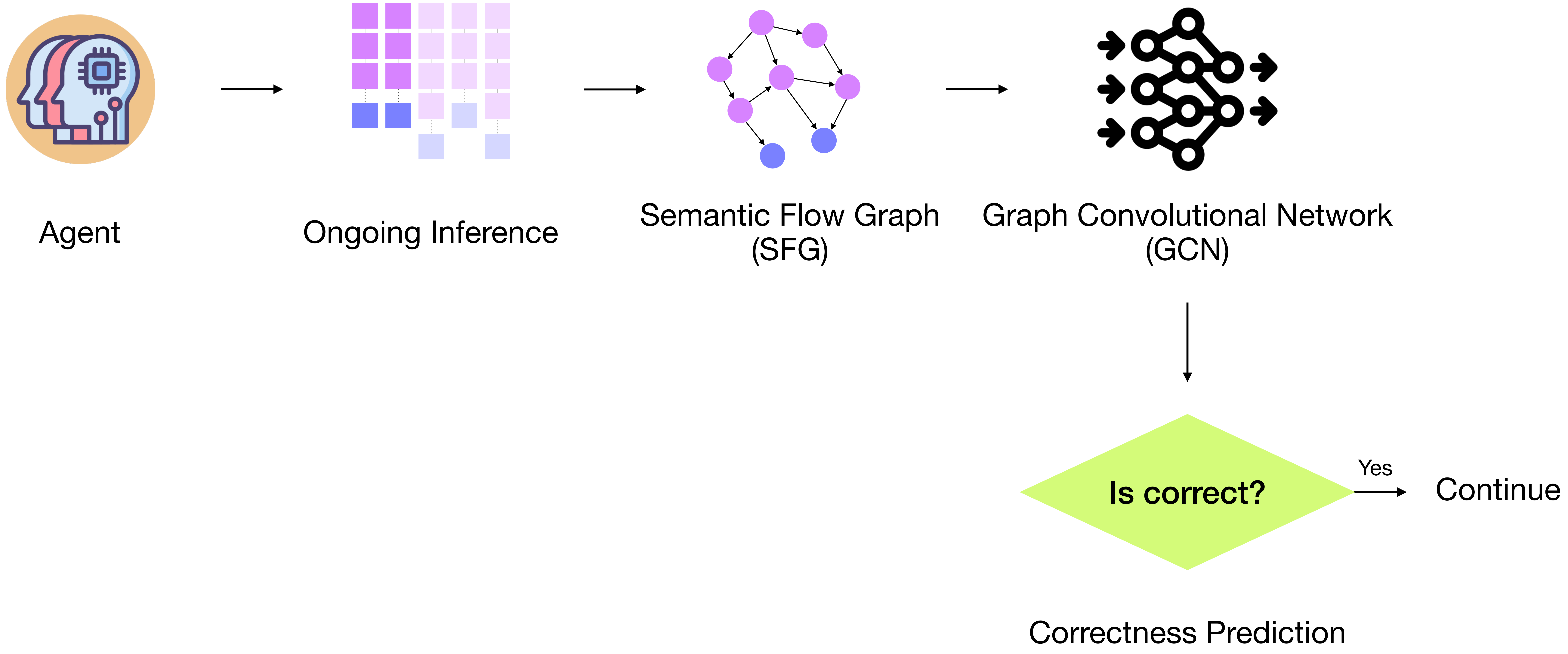

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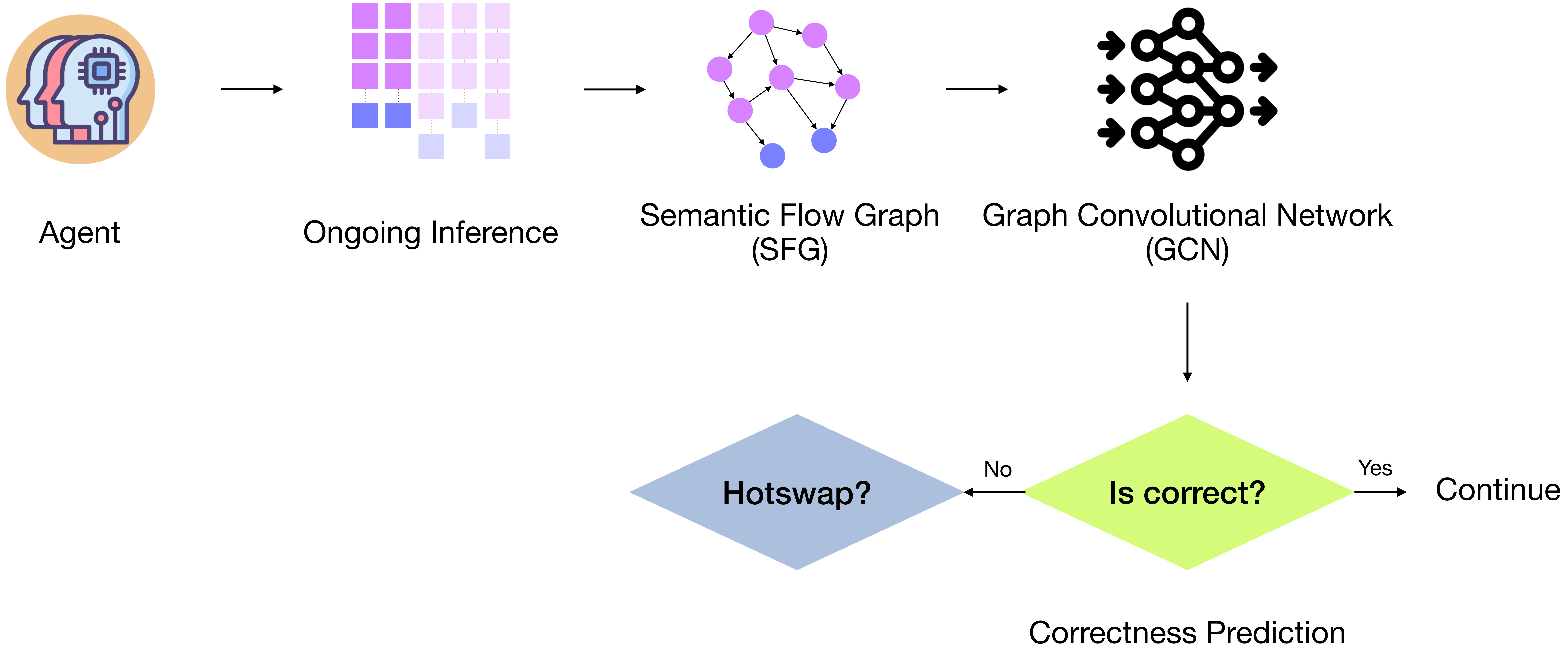

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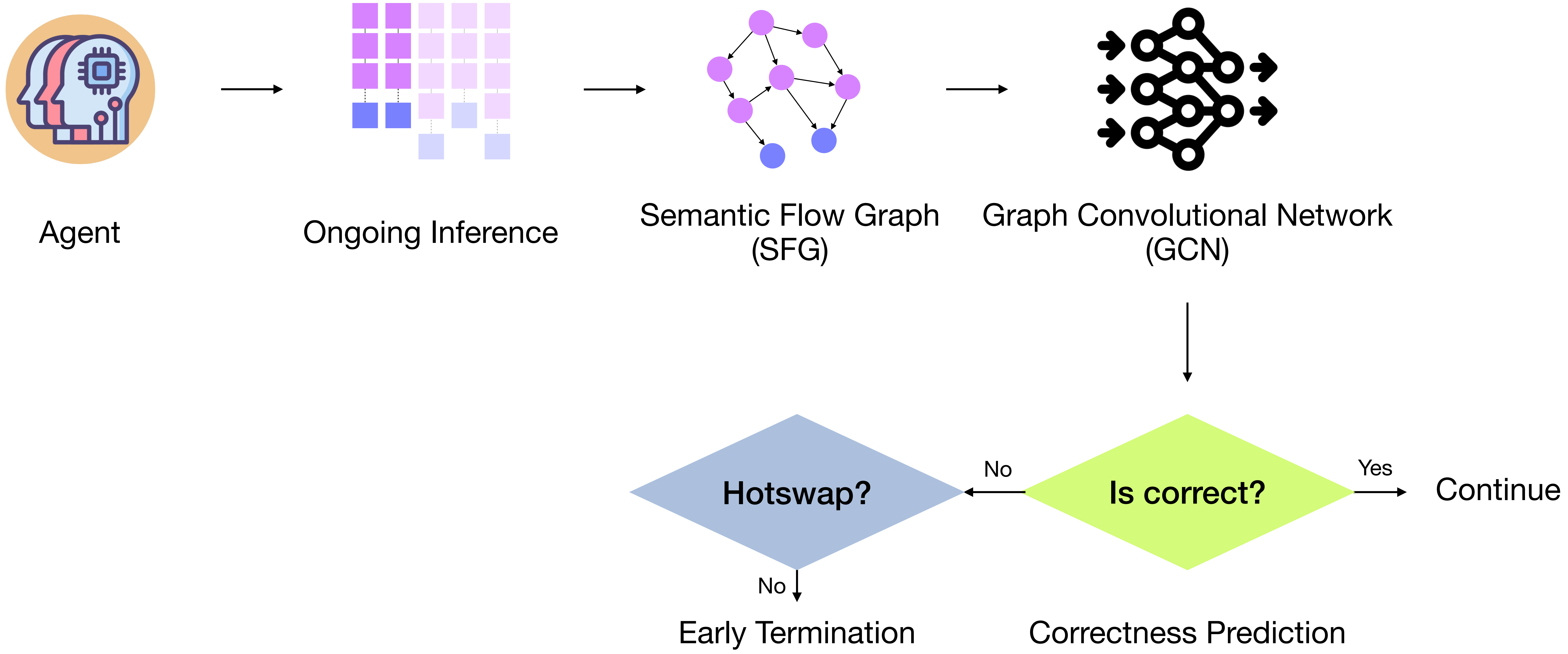

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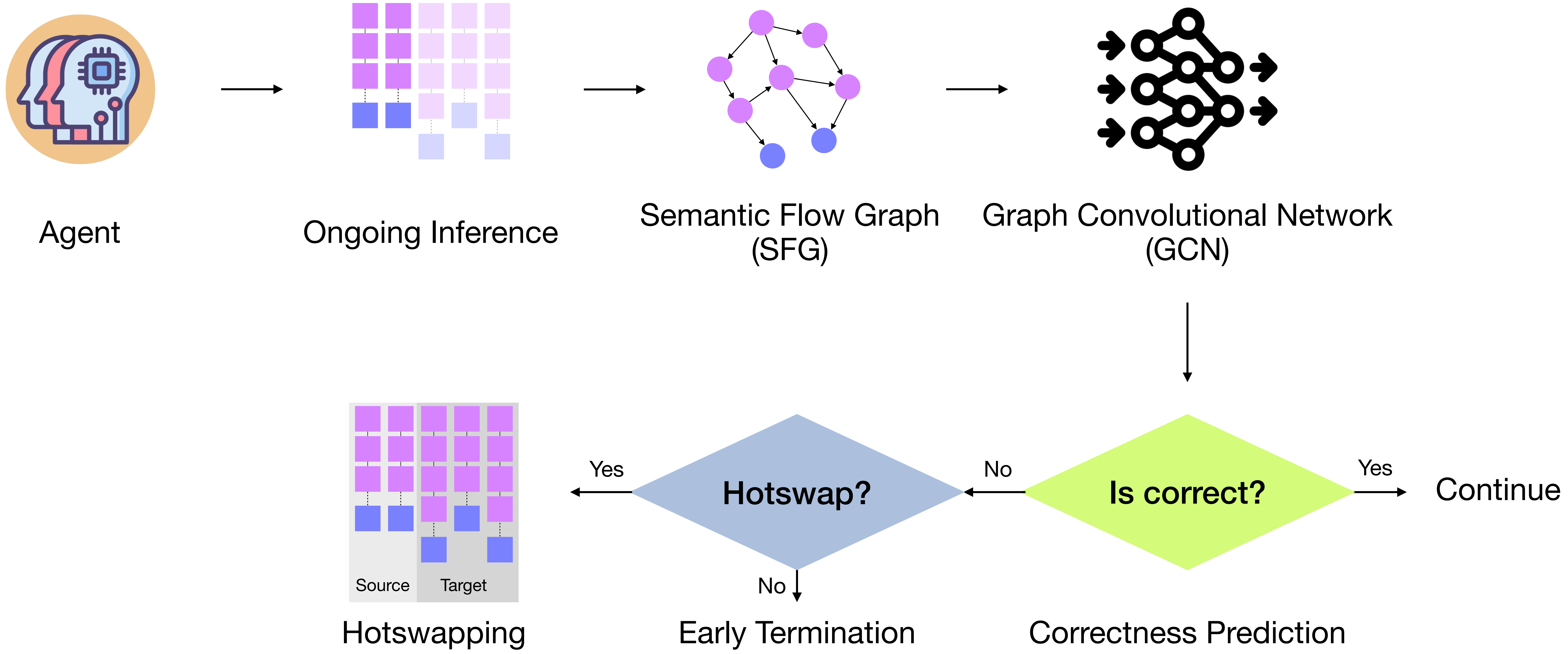
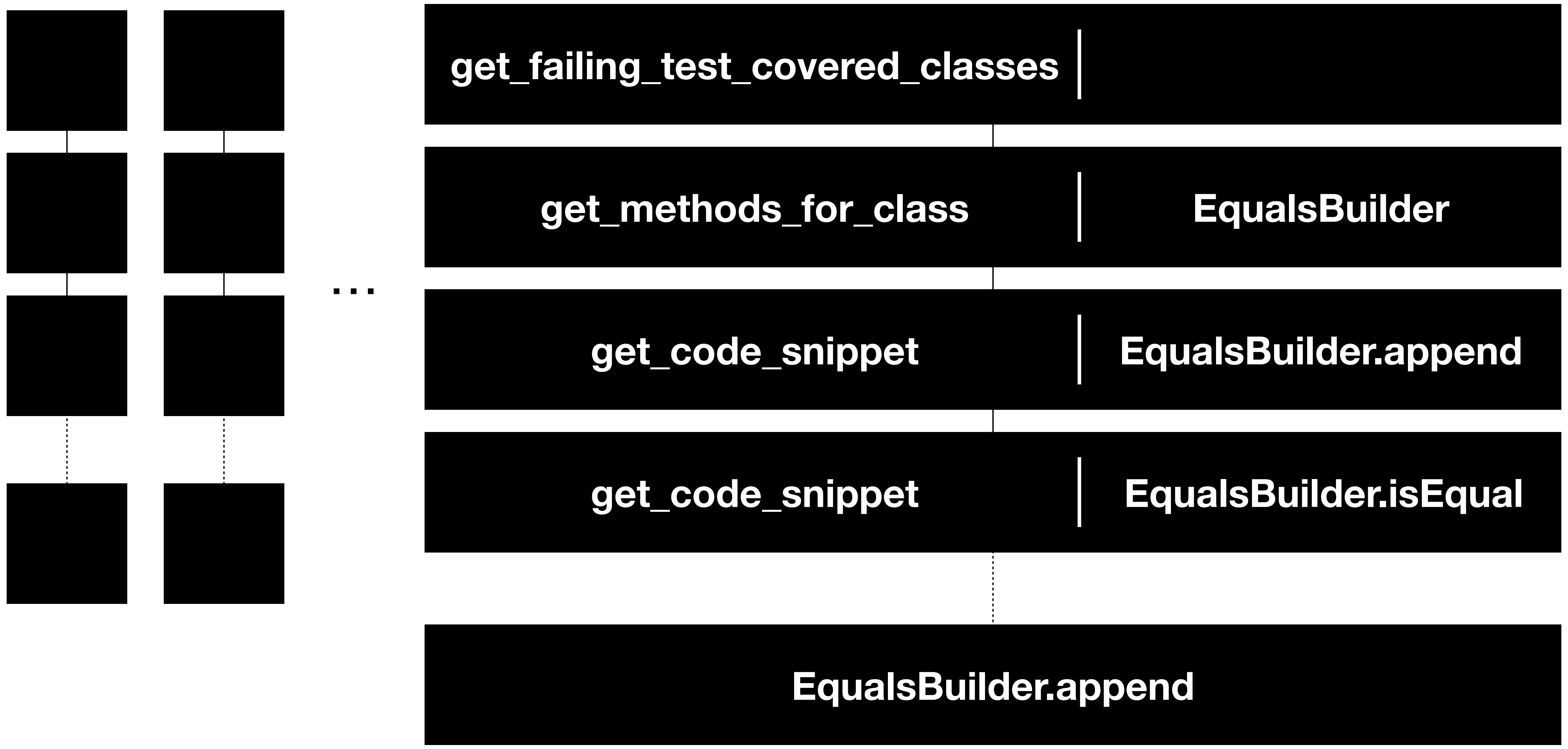


Semantic Flow Graph

AutoFL, AutoCodeRover

- AutoFL: FL agent, Autonomously navigates the repository using four external functions
- AutoCodeRover: APR agent, use only FL phase for self-consistency



Example Trajectories
(taken from AutoFL inferences)

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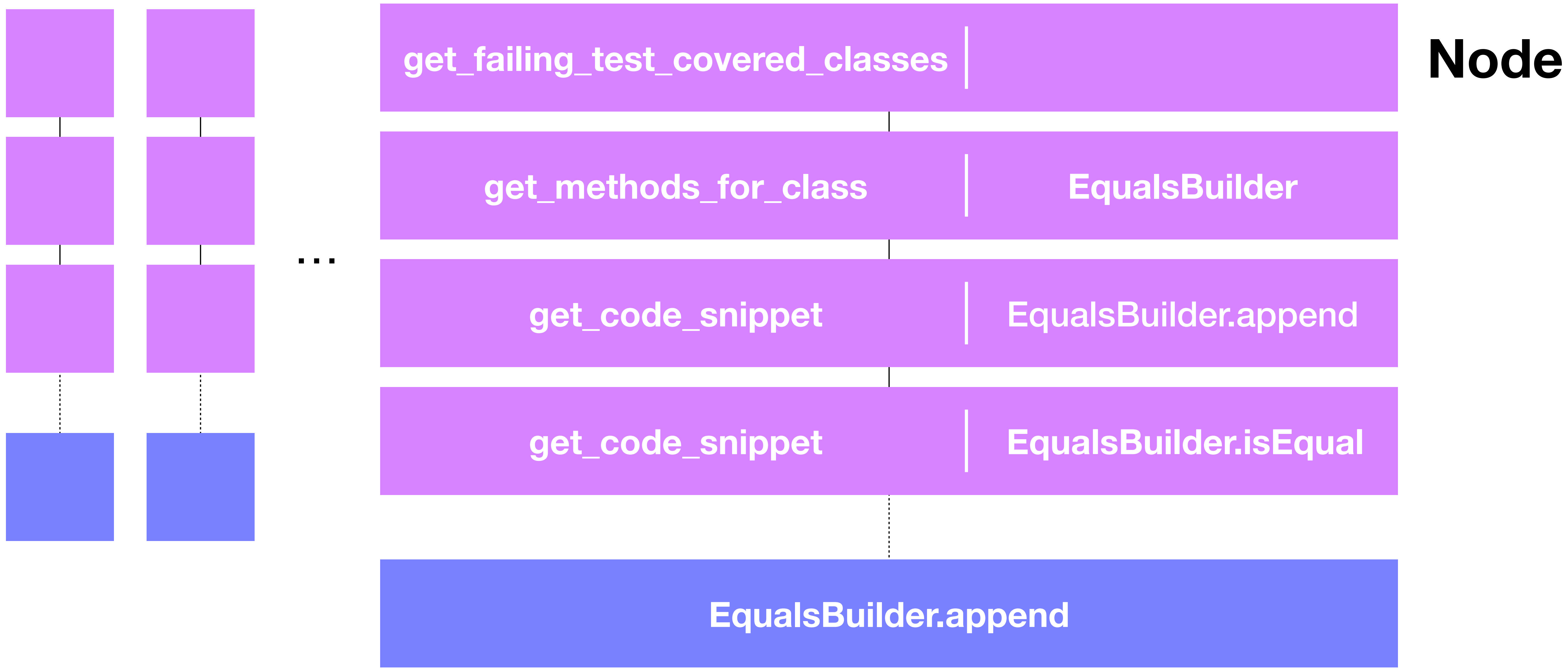

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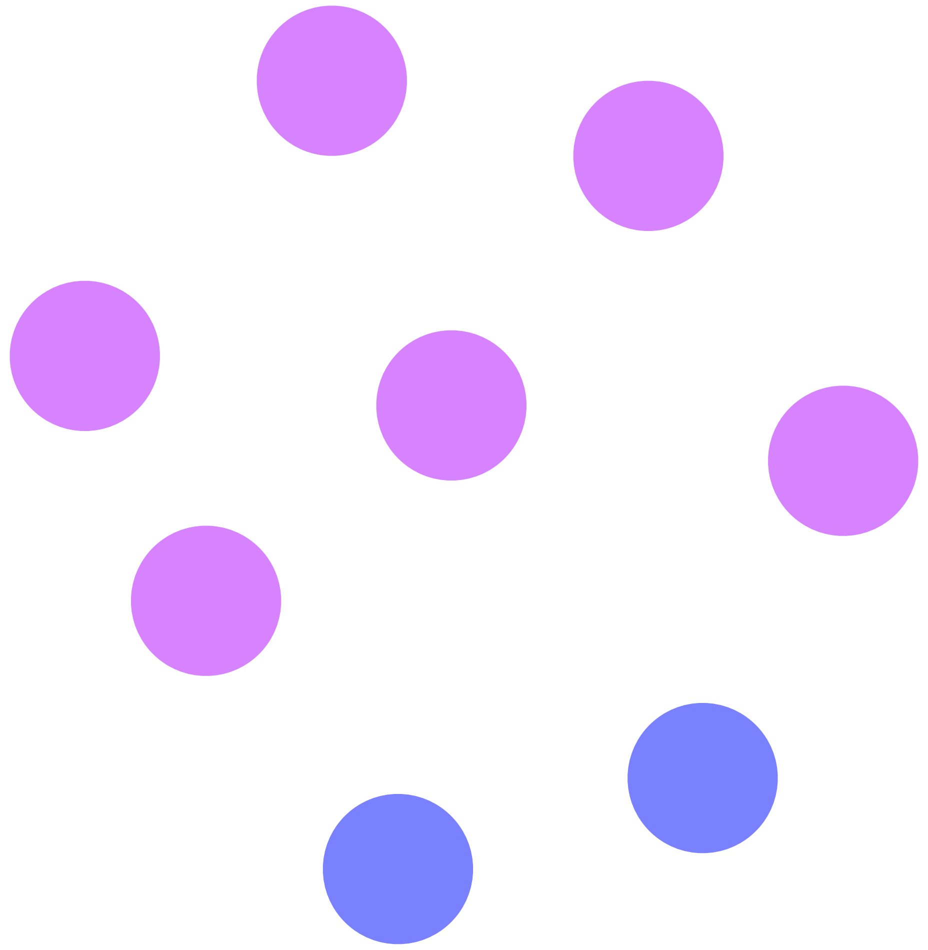

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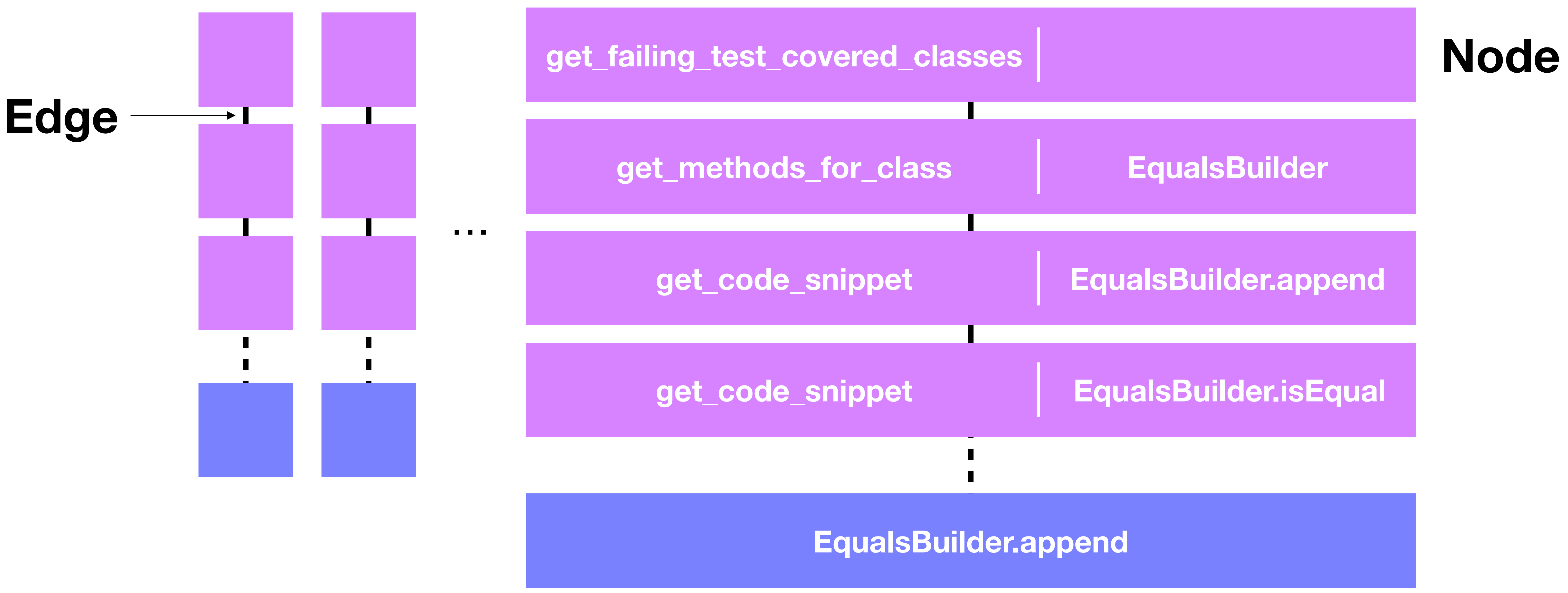

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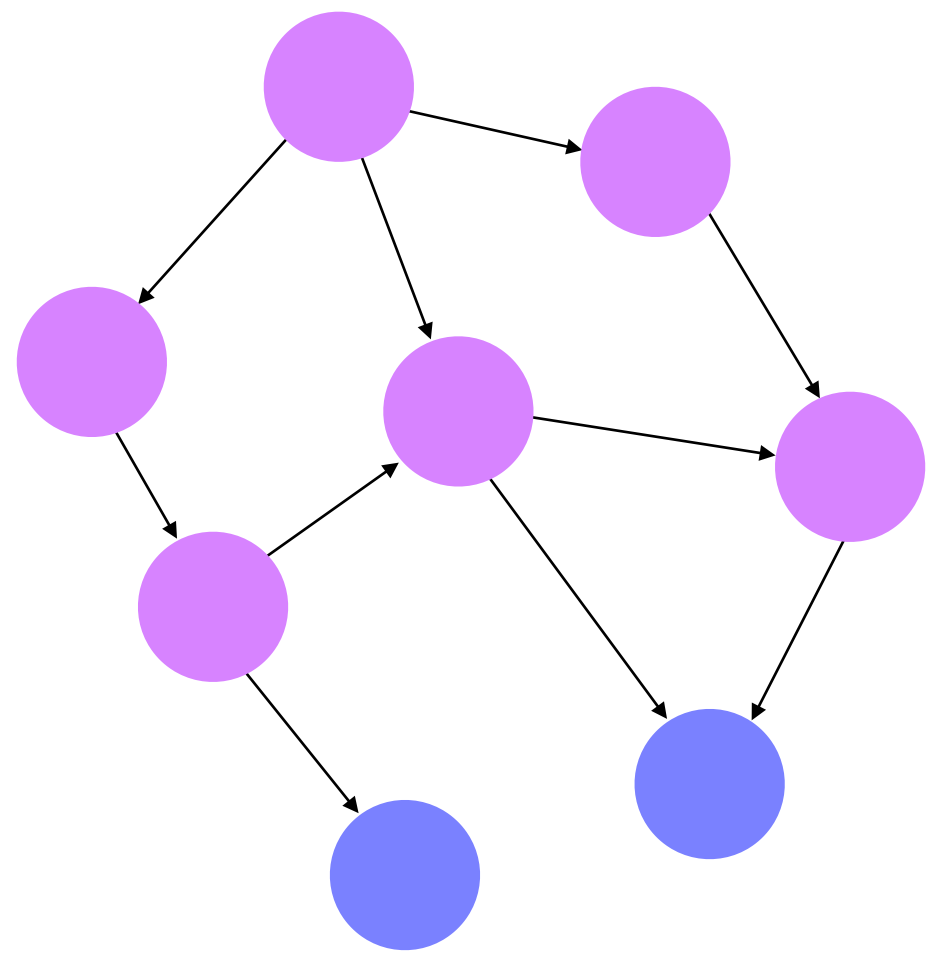

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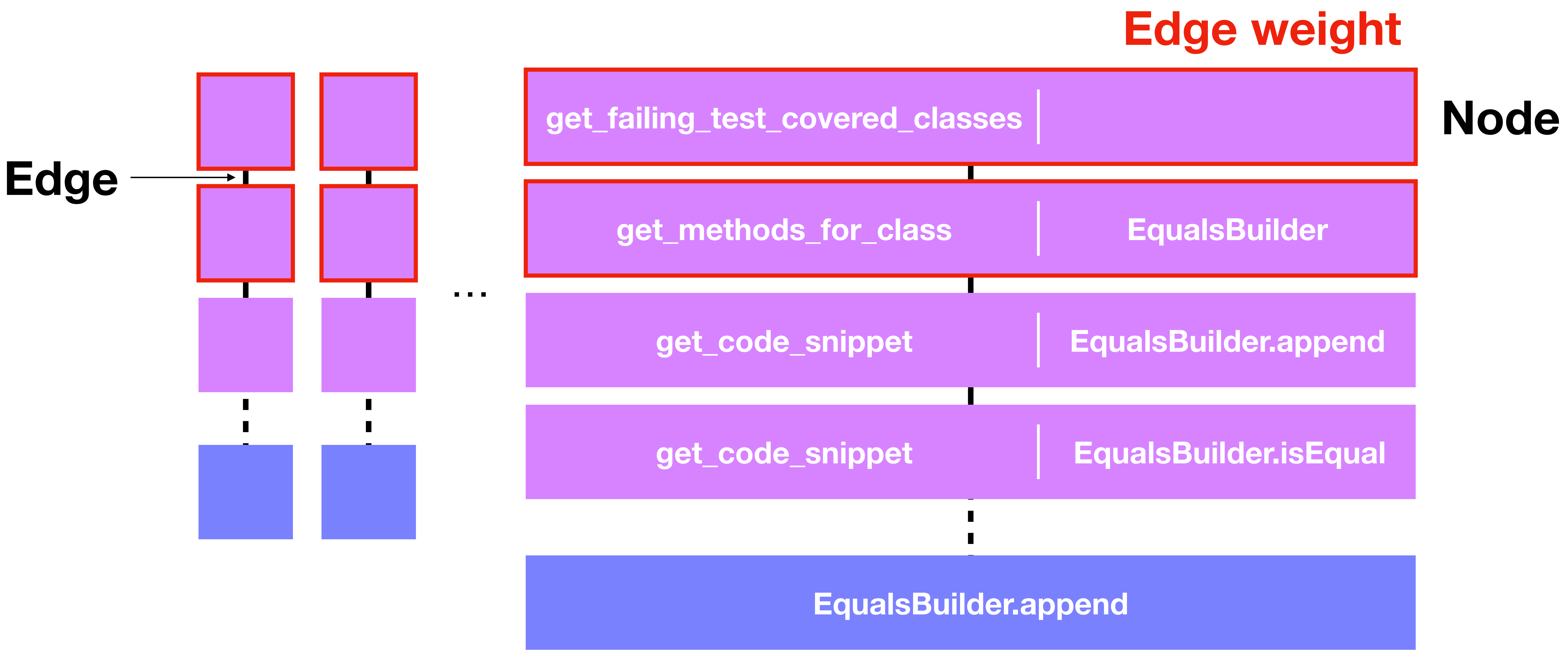

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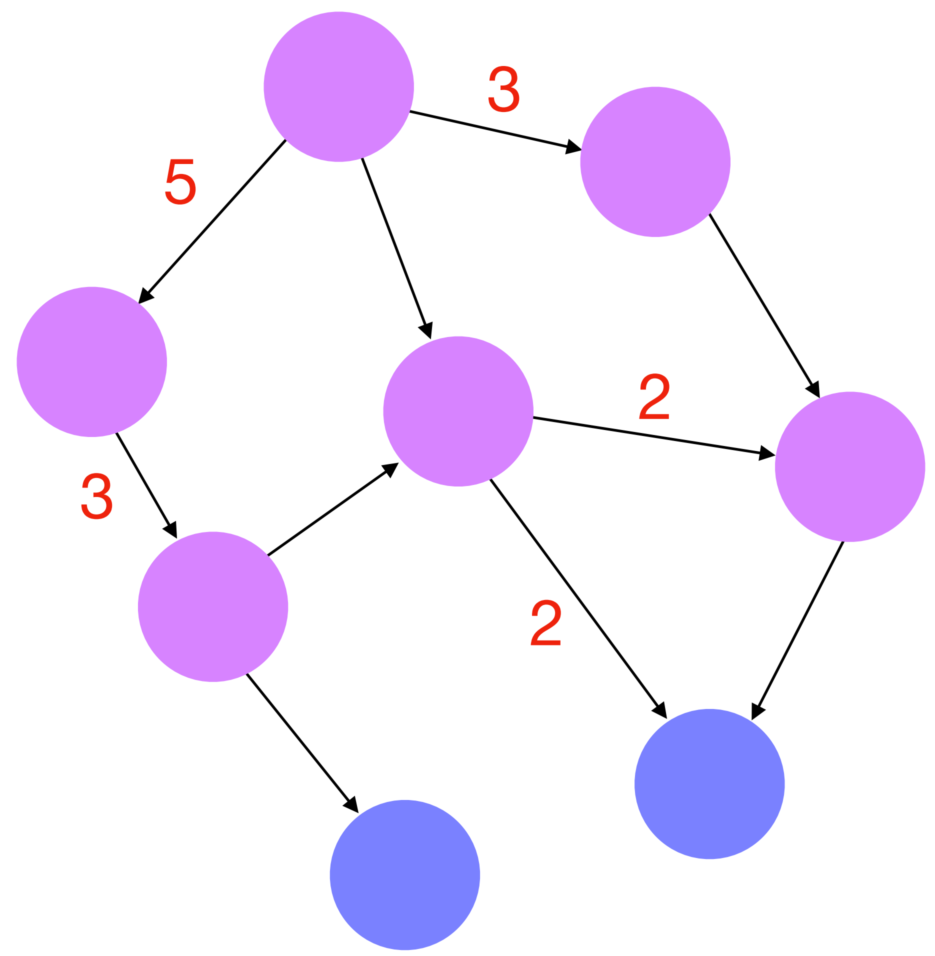
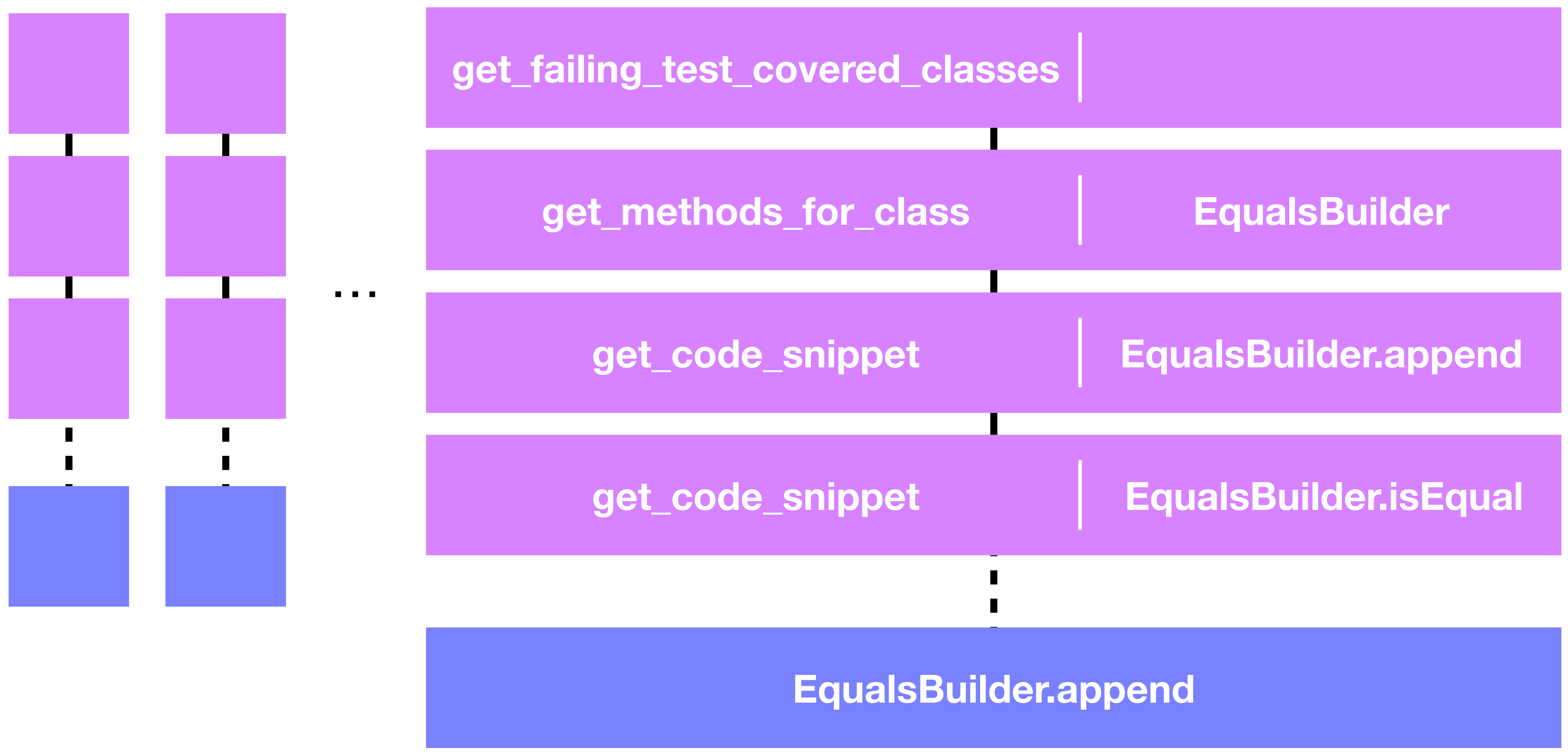


Semantic Flow Graph

Semantic Flow Graph

Node Embedding



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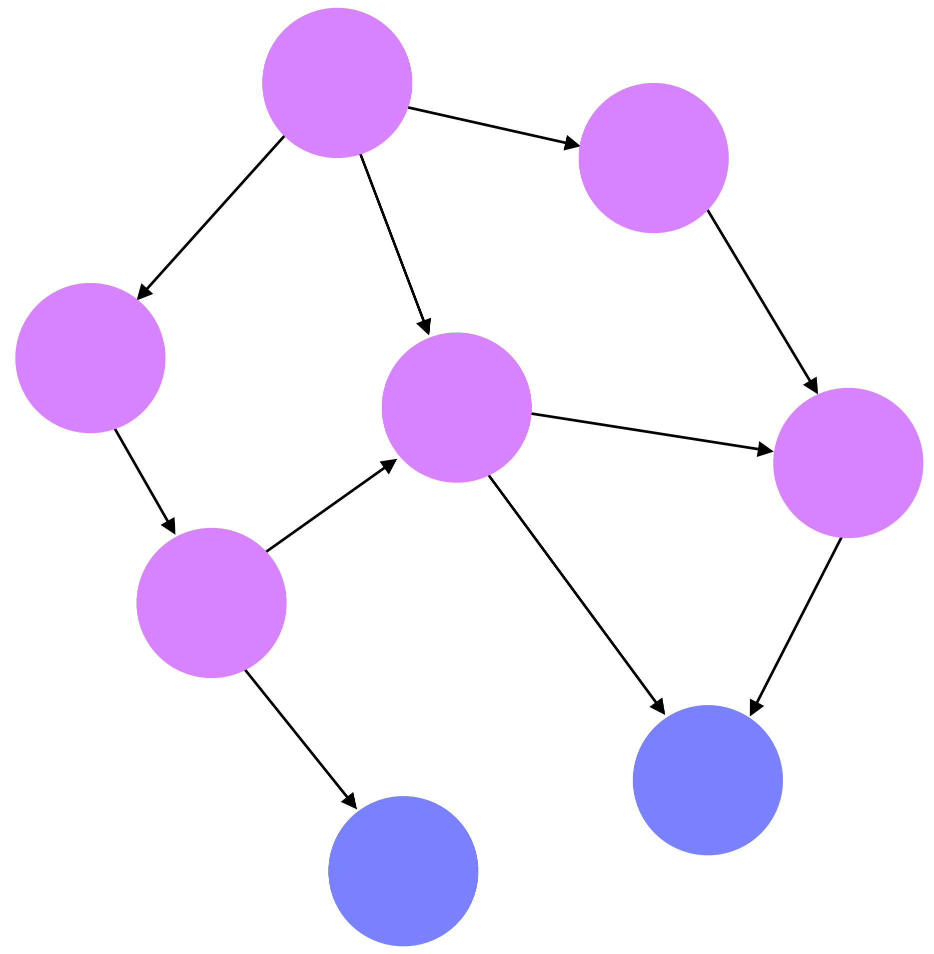

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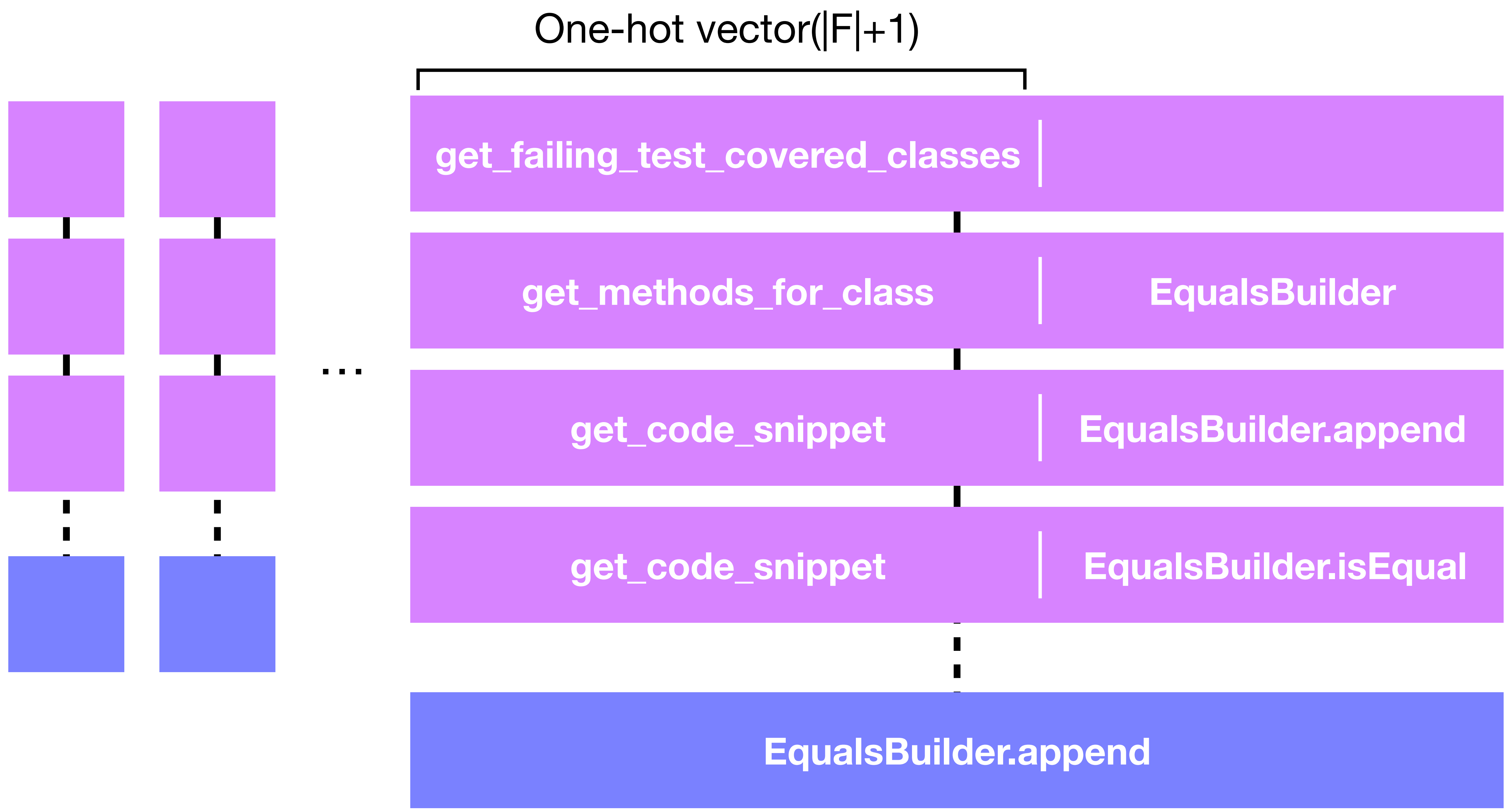

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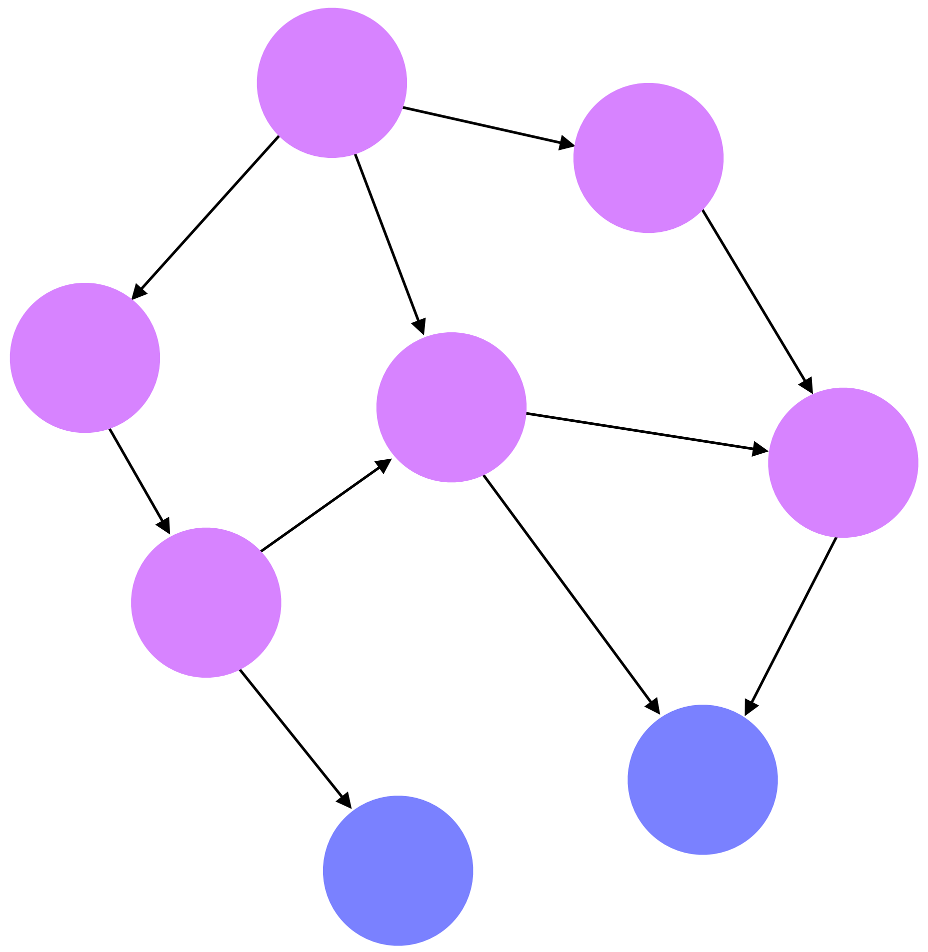

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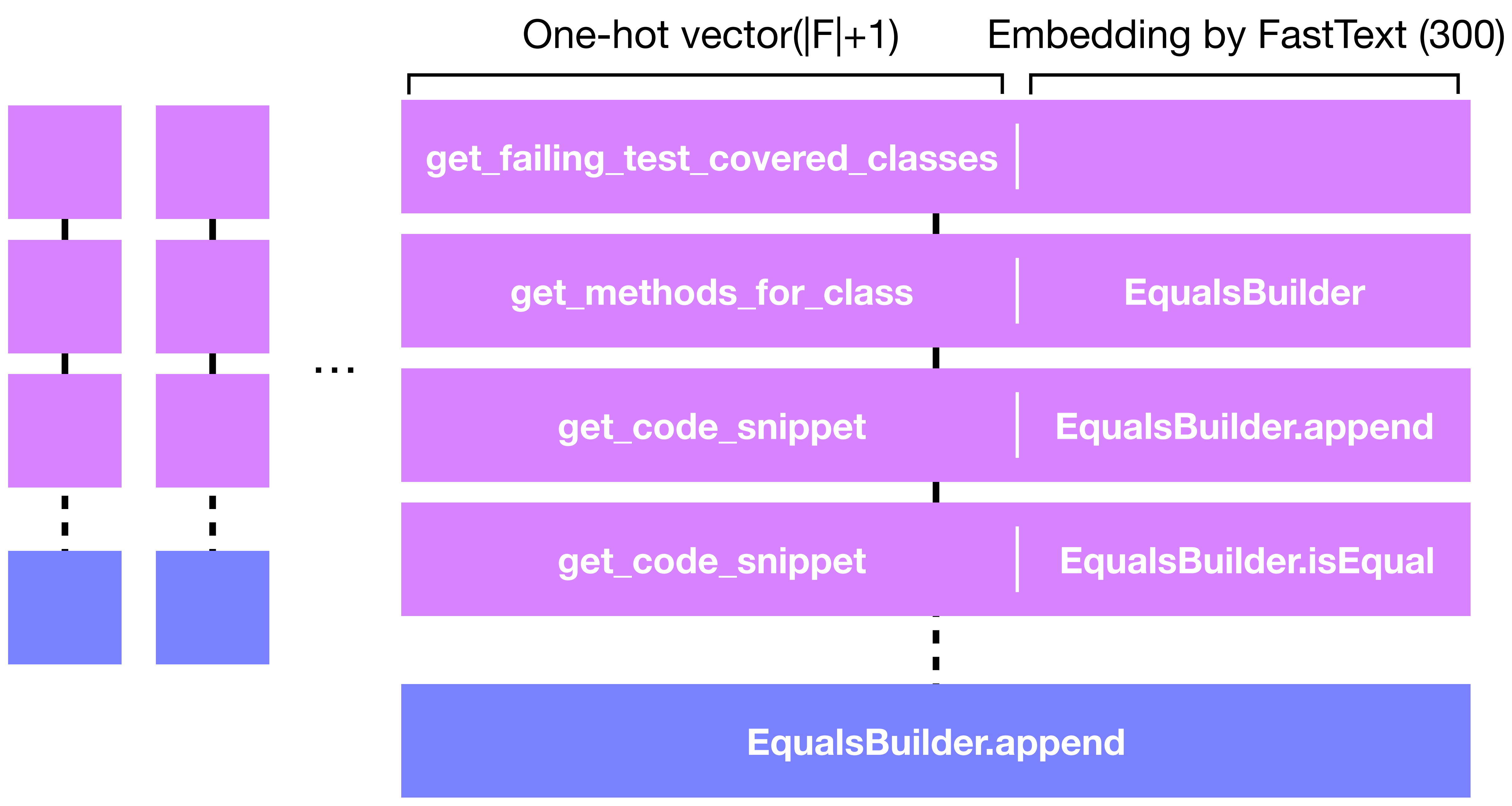

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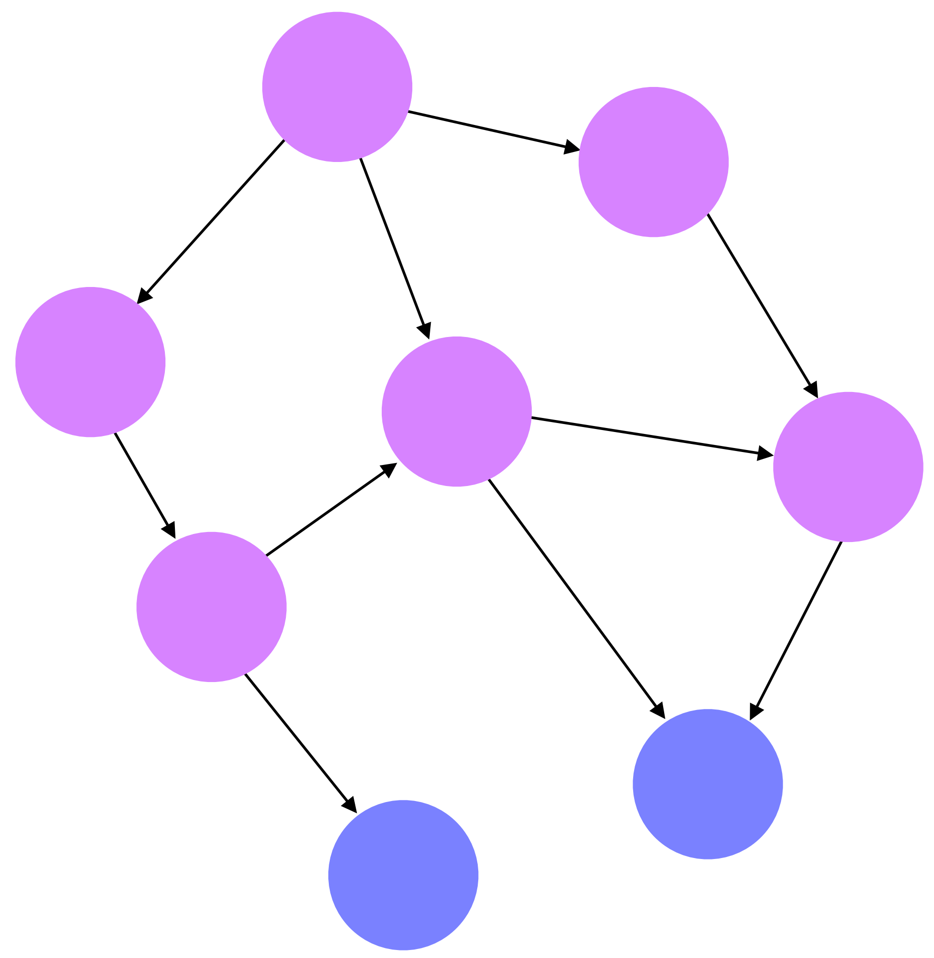

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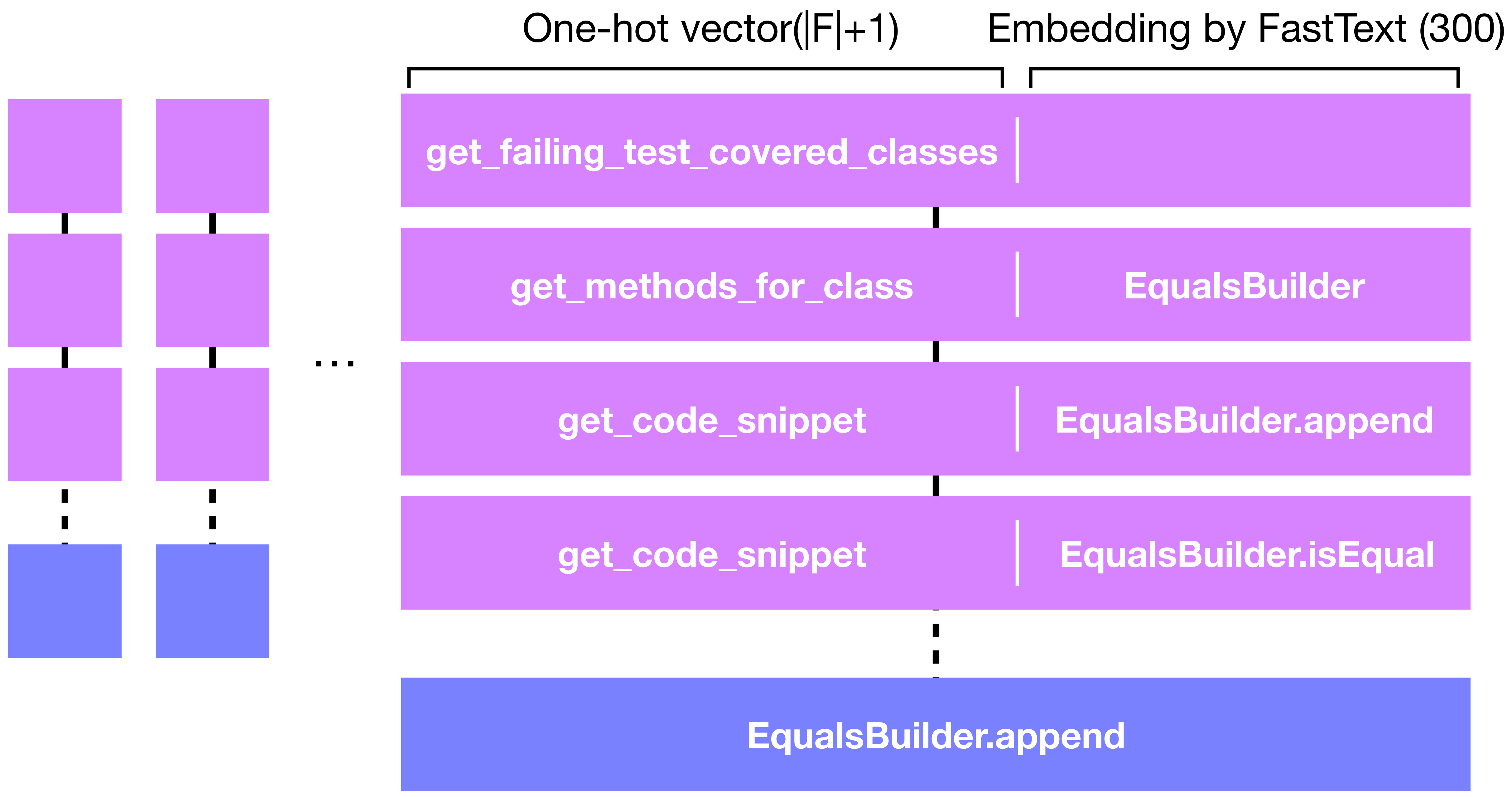

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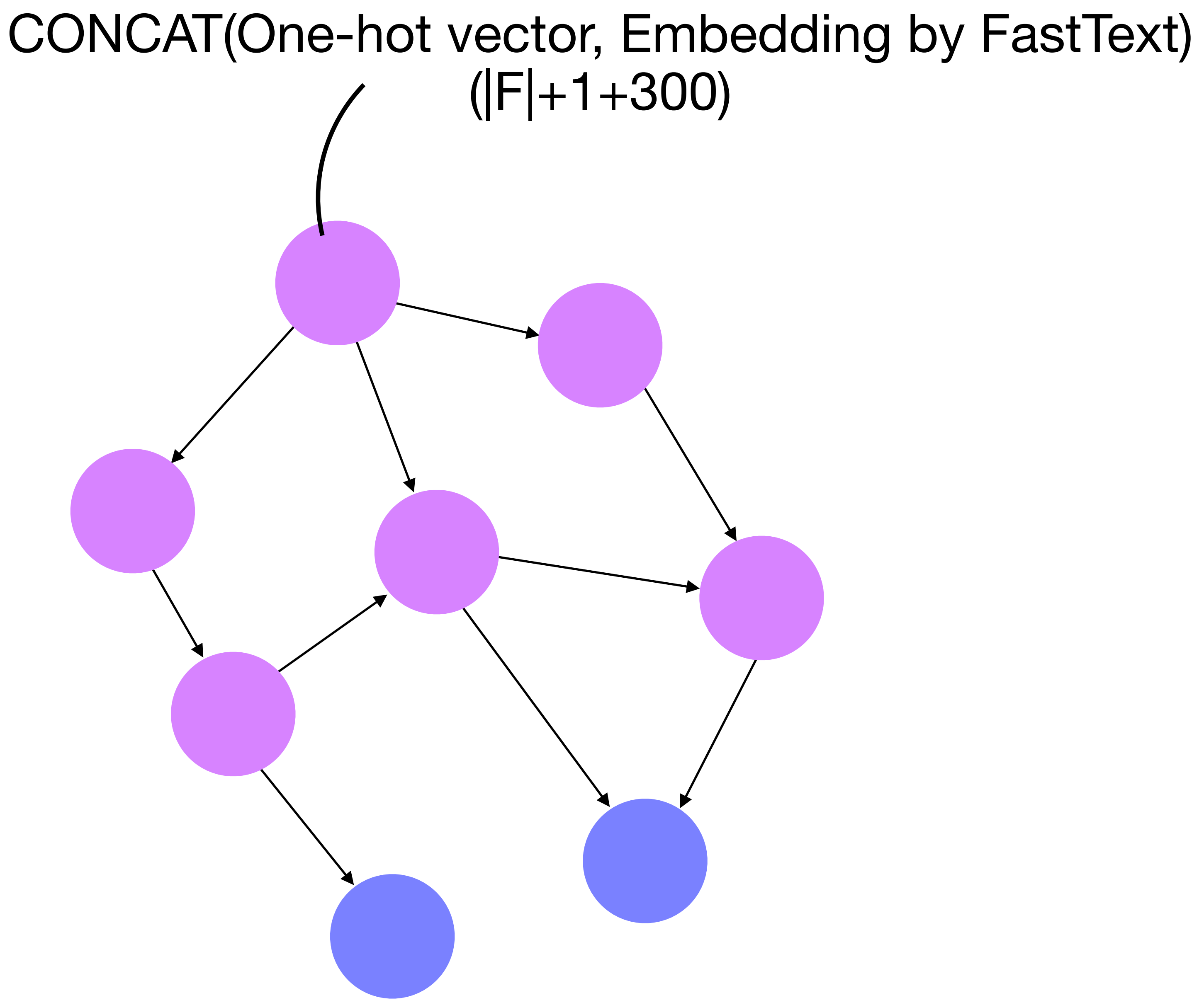
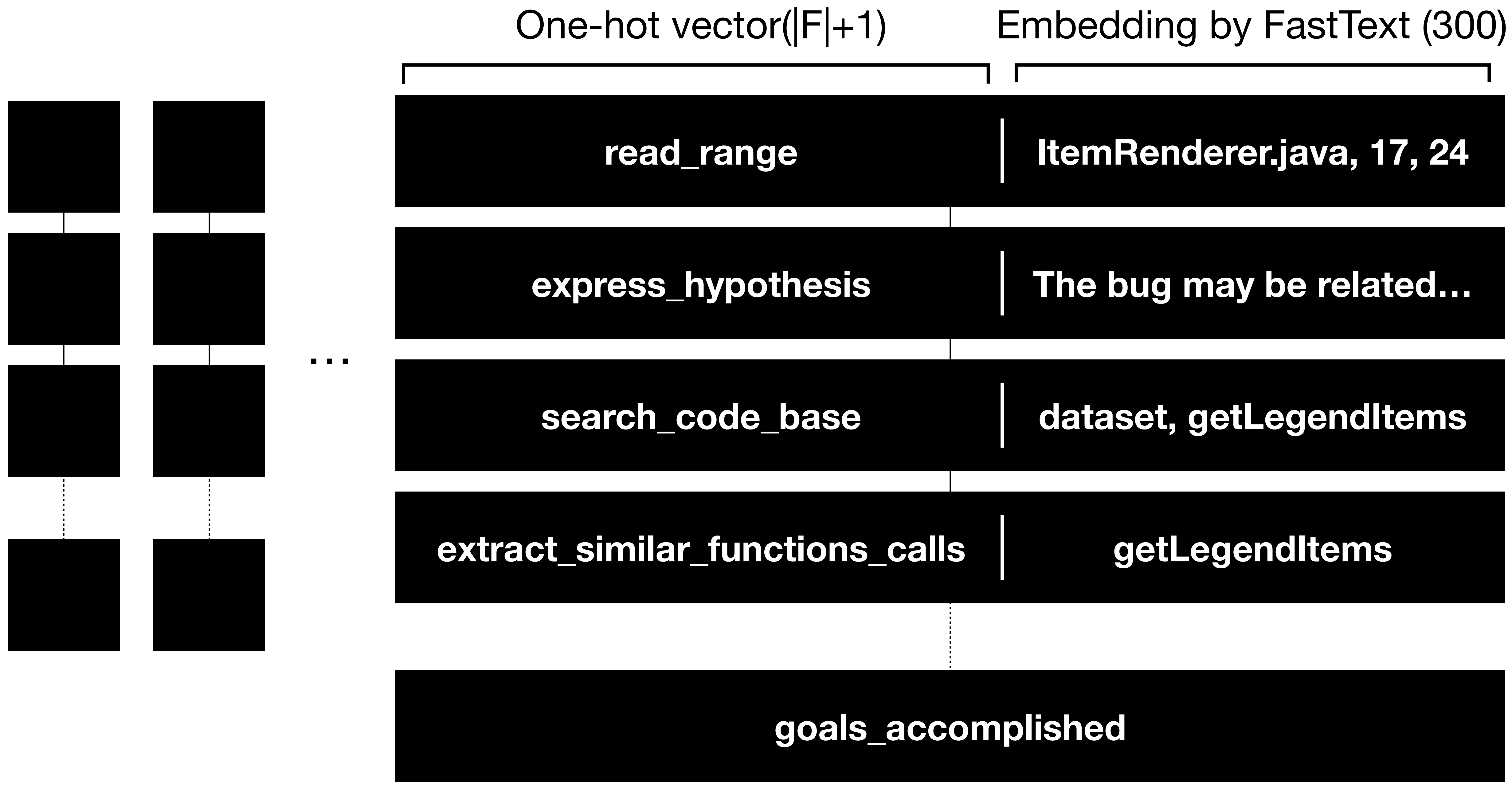


Semantic Flow Graph

Semantic Flow Graph

RepairAgent

- RepairAgent: APR agent, takes natural language and code patch as arguments



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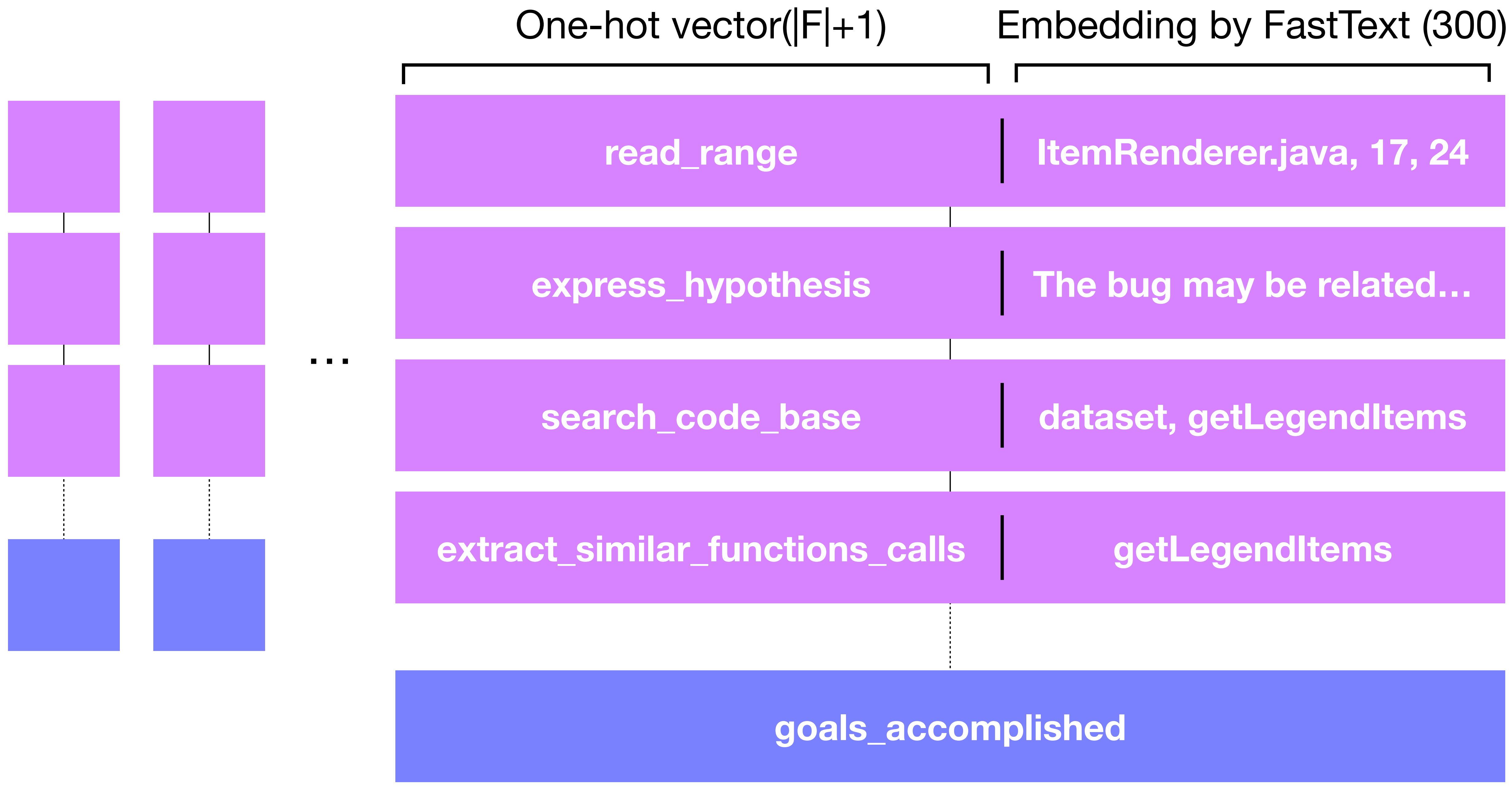

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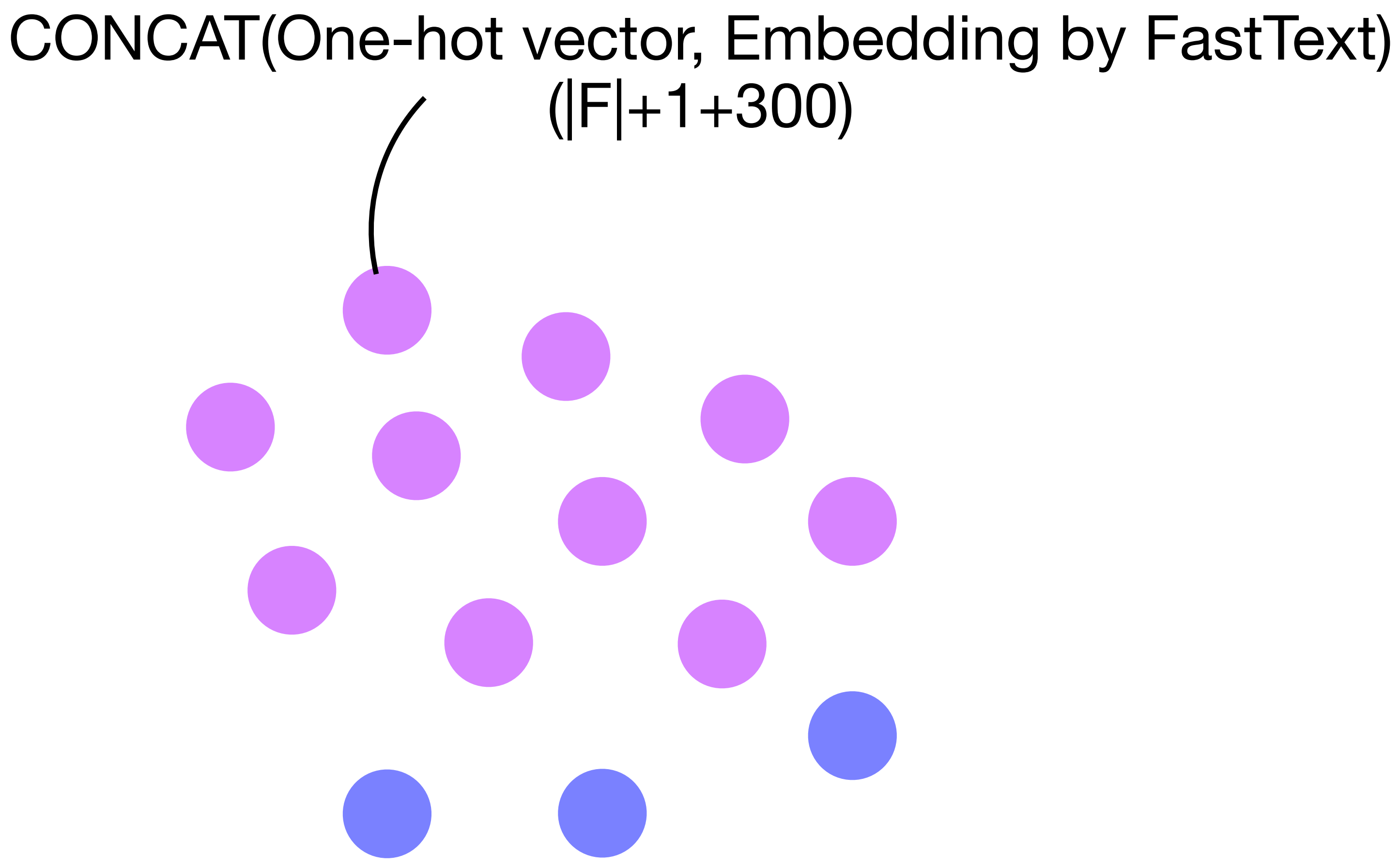

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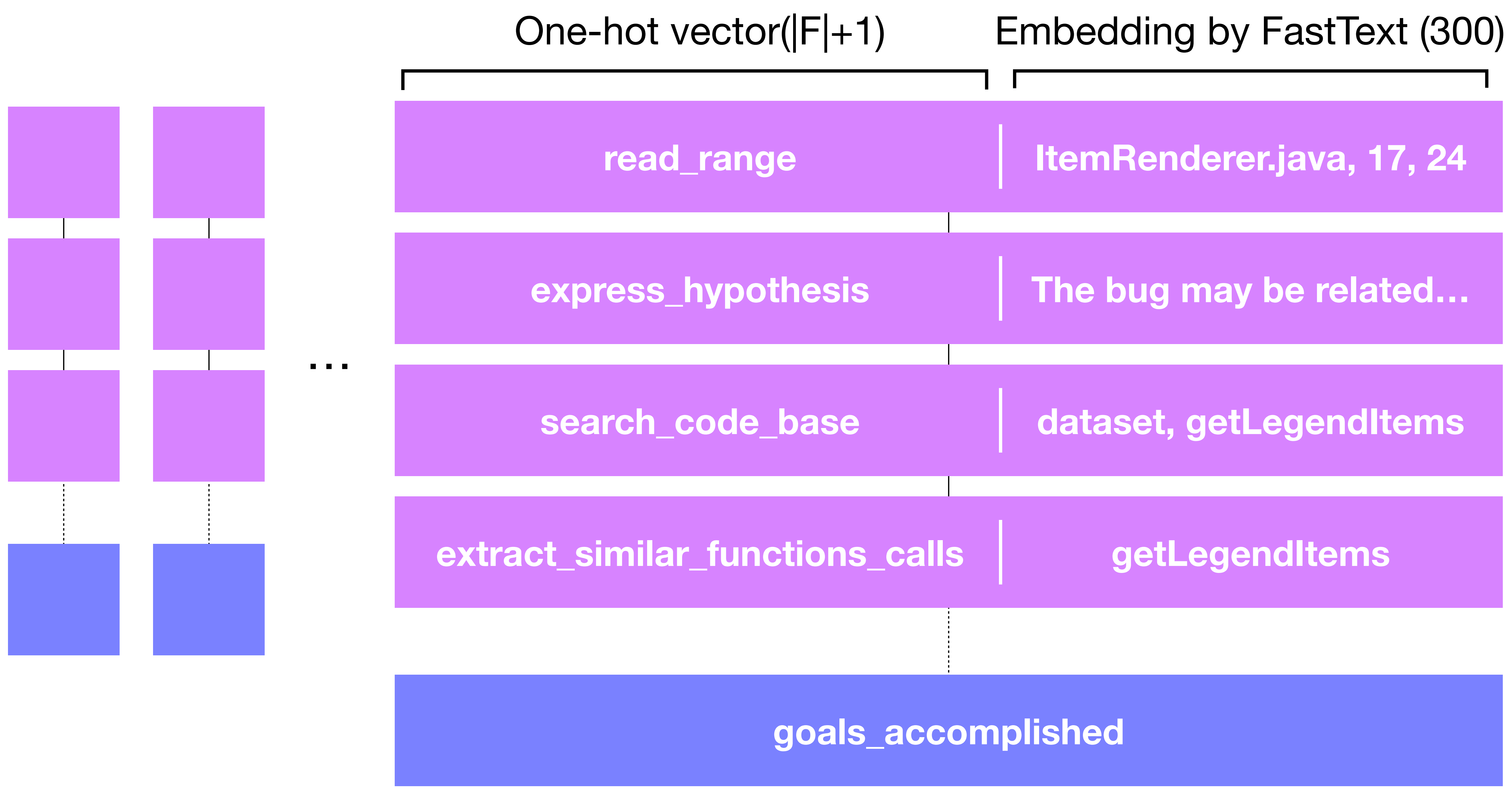

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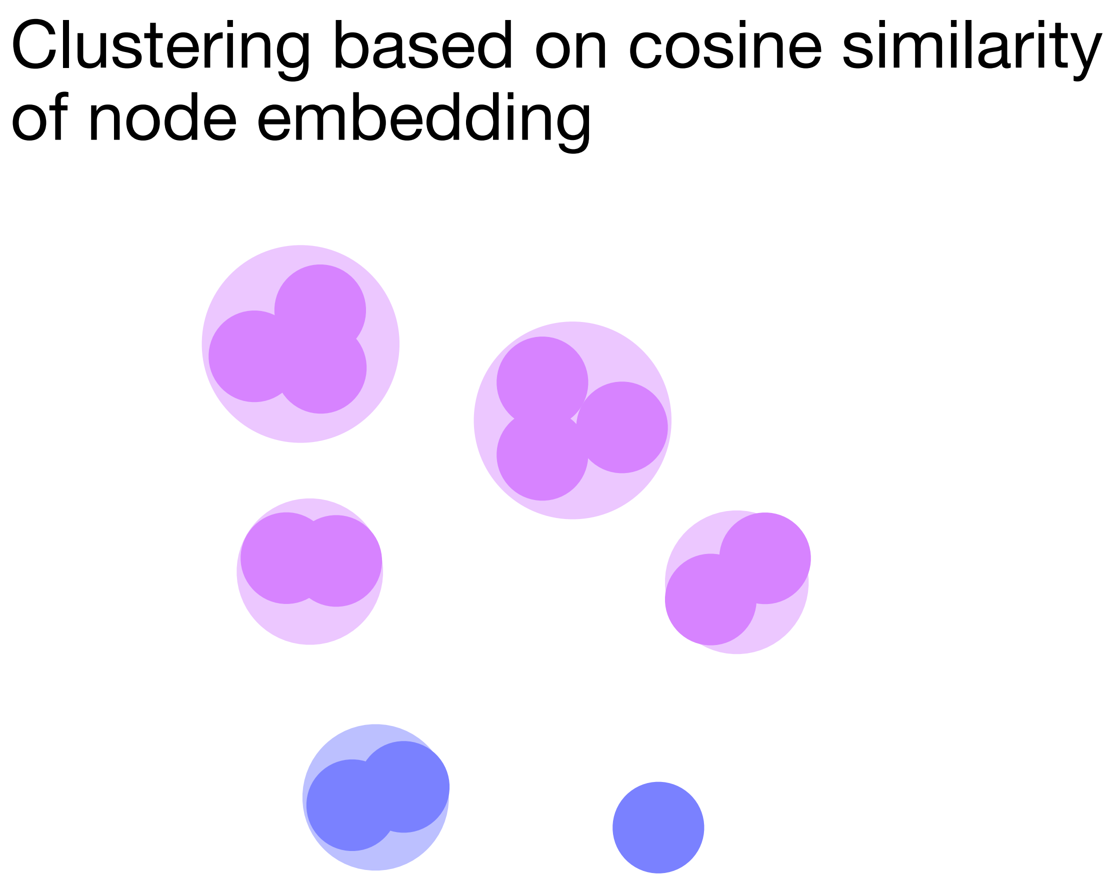

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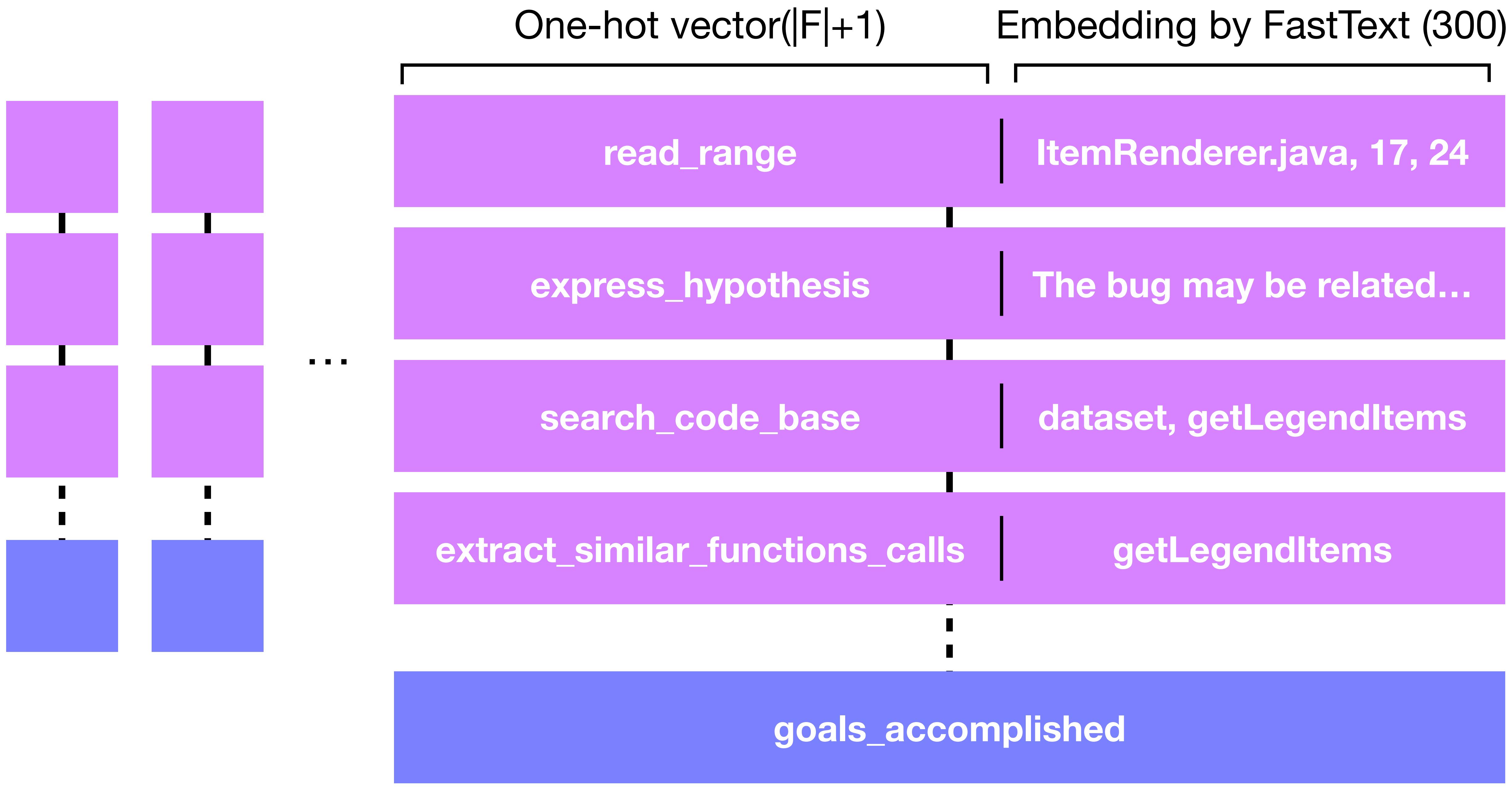
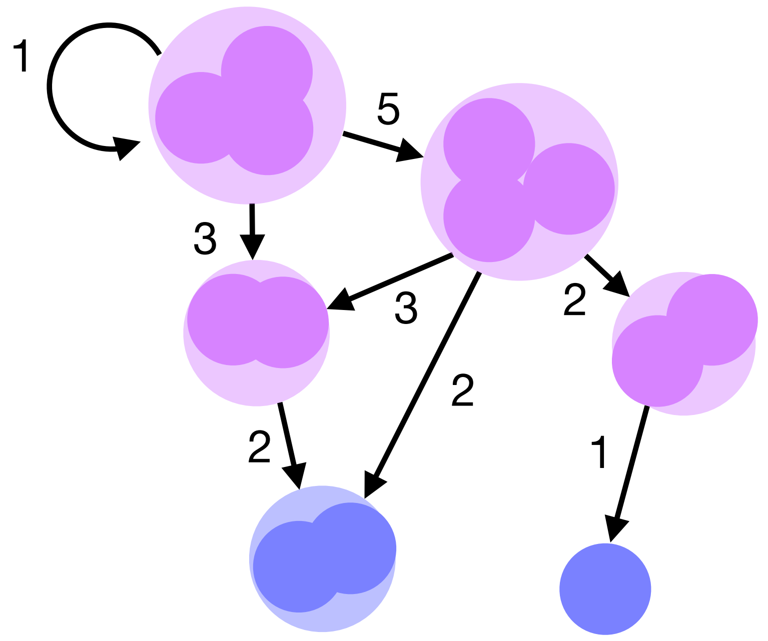


Example Trajectories
(Taken from RepairAgent inferences)

Add edges and weights based on sequences of trajectories

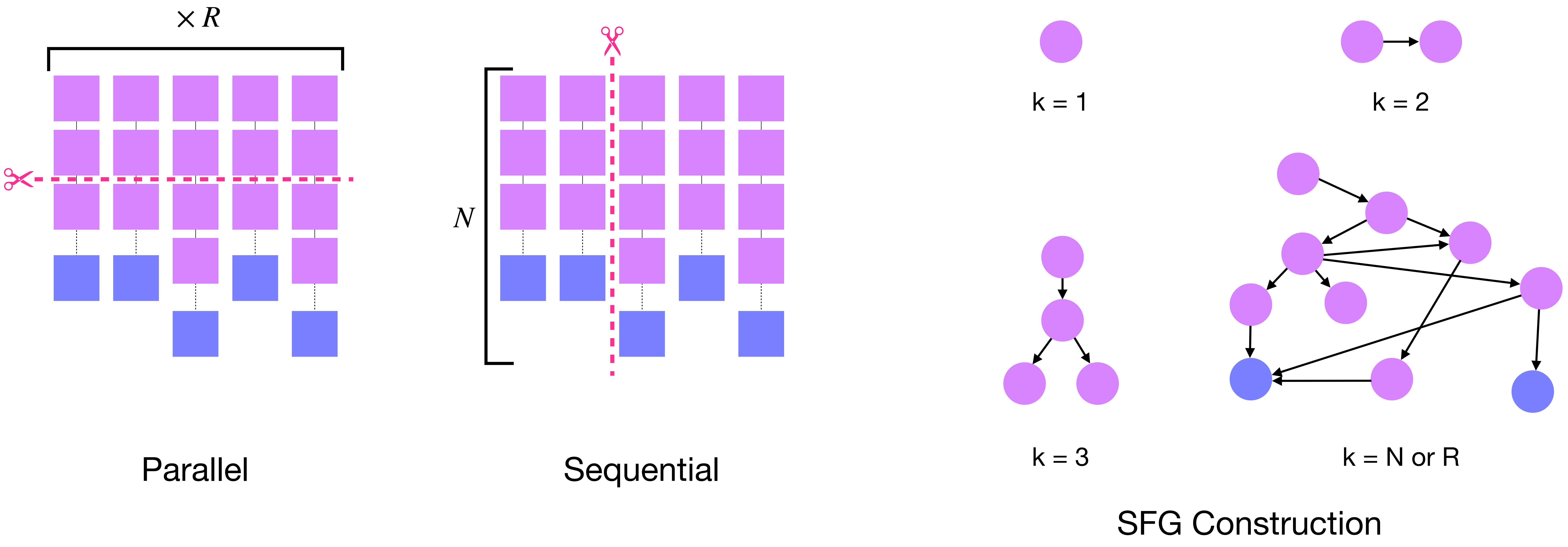


Semantic Flow Graph

Correctness Prediction

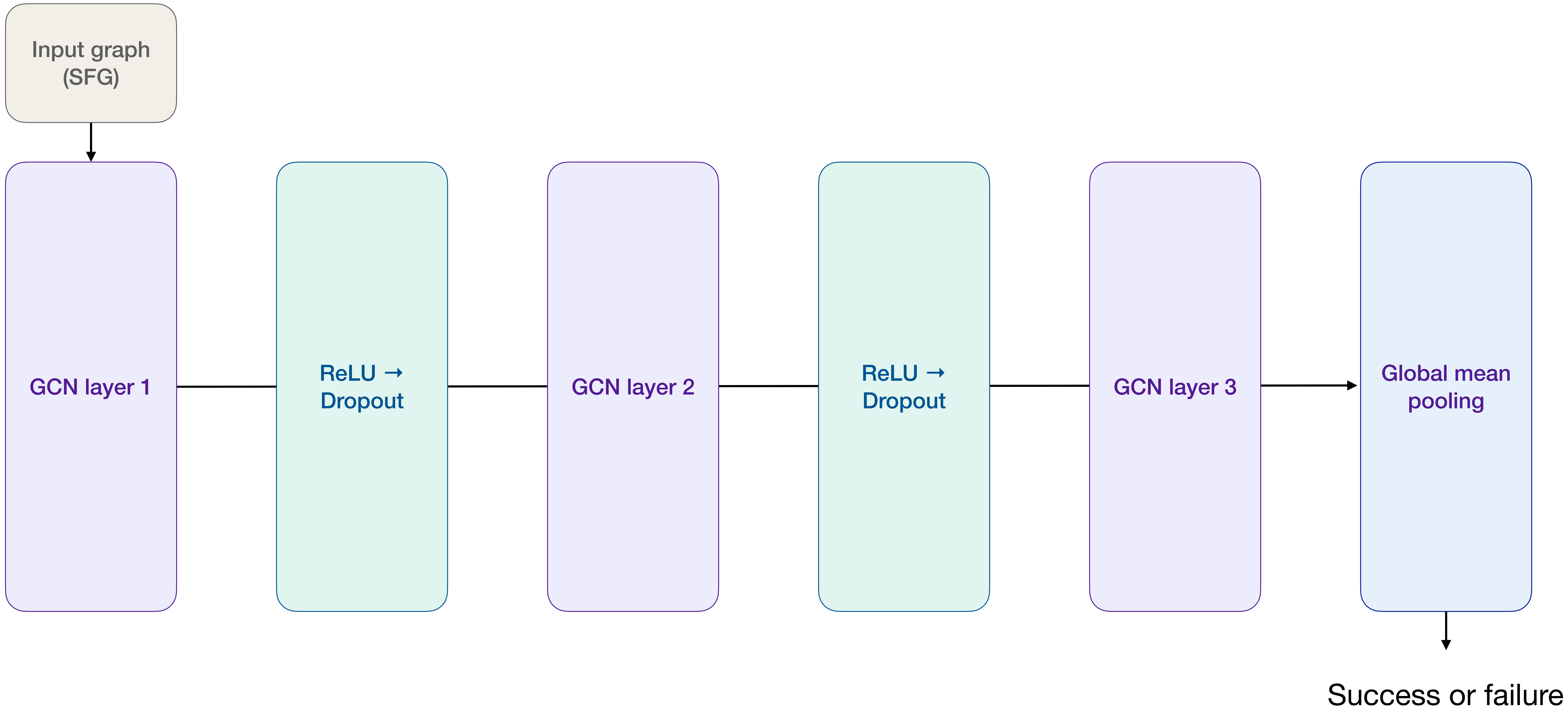
Trajectory Truncation

- **Truncate** completed trajectories at various k to build GCN training dataset



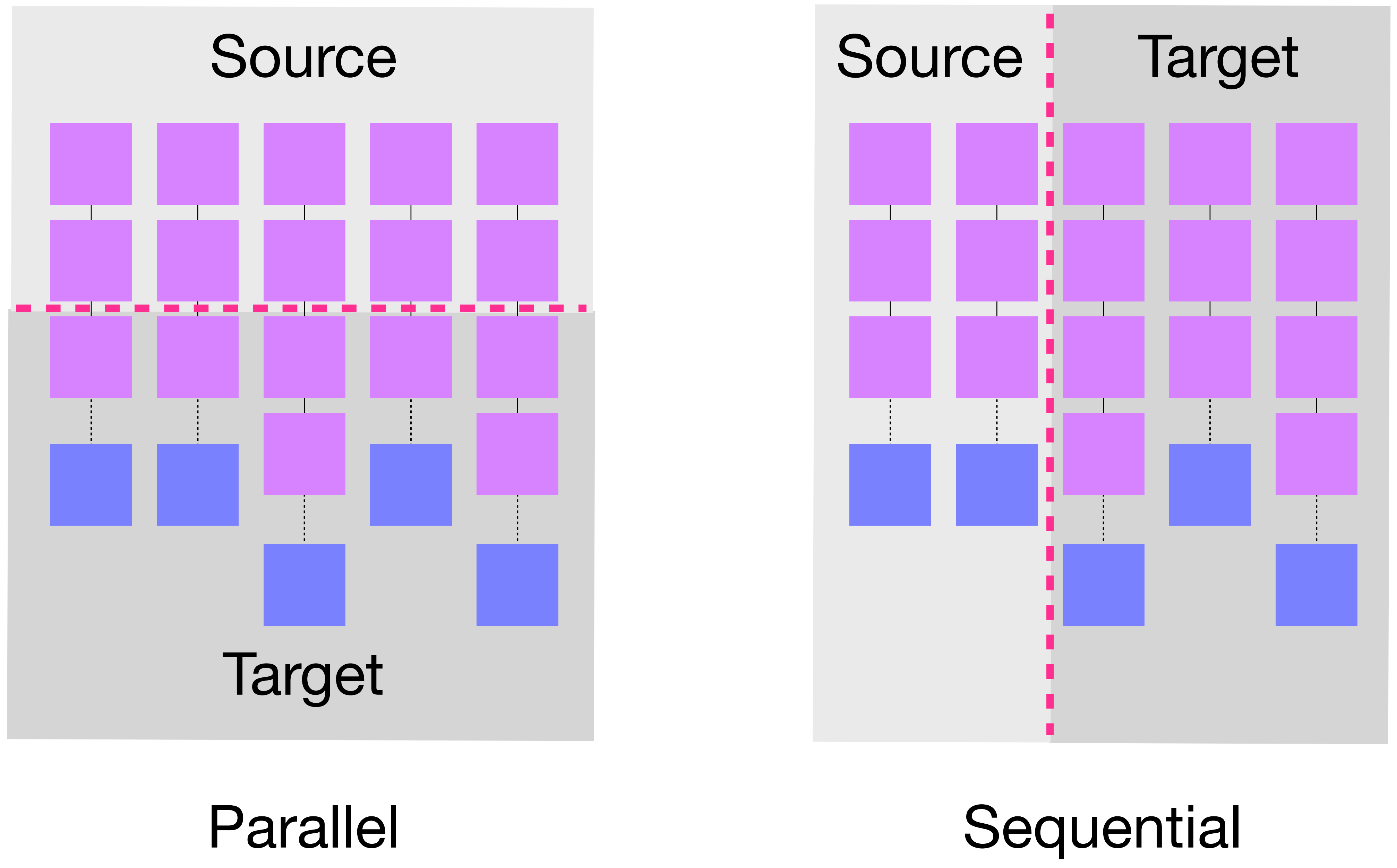
Correctness Prediction

Graph Convolutional Network



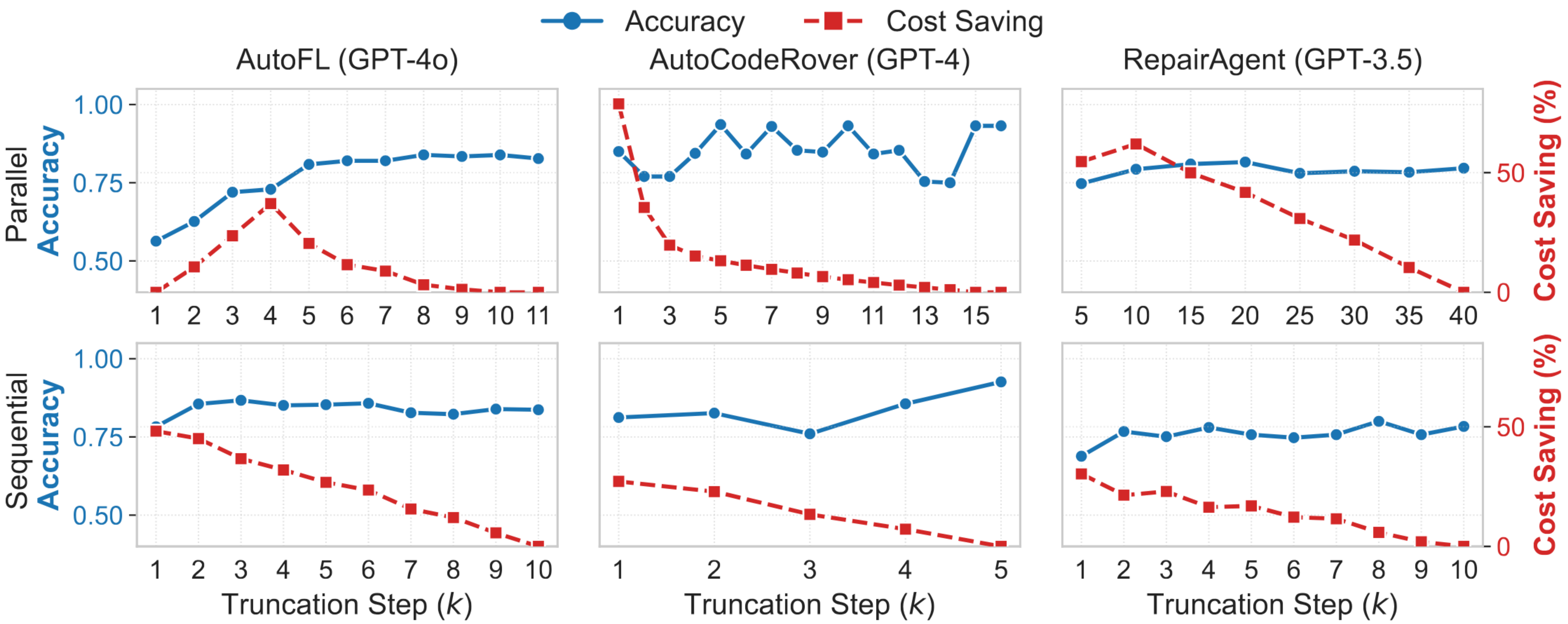
Hotswapping

- Source model: initial inference model; cheaper, less capable
  - e.g. Llama-3-8B
- Target model: swapped in when failure is predicted; more expensive, more capable
  - e.g. GPT-4o



Results

RQ2. Effectiveness of Early Termination



Trade-off analysis between Prediction Accuracy and Cost Saving

Results

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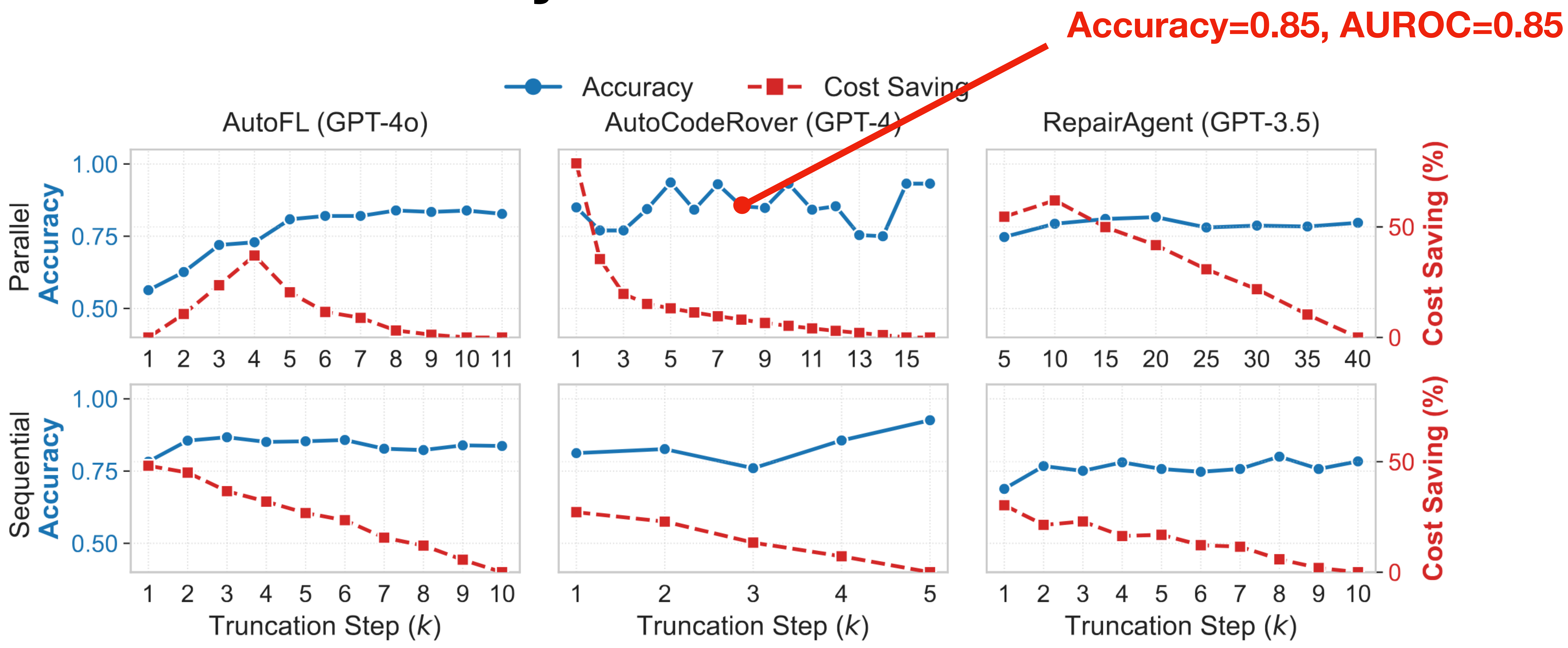

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RQ3. Hotswap Effectiveness

- Parallel hot swapping at a $k = \lceil N/2 \rceil$ only, due to significant financial cost

Agent	# Bugs	Method	Cost ($) (Ratio)	$LC(1)$ (Ratio)	$LC(3)$ (Ratio)	$LC(5)$ (Ratio)
AutoFL	428	Source (Llama-3)	2.63 (0.91%)	120 (52.17%)	172 (58.50%)	189 (61.36%)
		Hotswap	69.01 (23.90%)	171 (74.35%)	239 (81.29%)	255 (82.79%)
		Target (GPT-4o)	288.80 (100.00%)	230 (100.00%)	294 (100.00%)	308 (100.00%)
AutoCodeRover	500	Source (Mixtral)	129.52 (24.38%)	111 (53.88%)	132 (50.38%)	142 (48.63%)
		Hotswap	442.44 (83.28%)	160 (77.67%)	212 (80.92%)	239 (81.85%)
		Target (GPT-4)	531.29 (100.00%)	206 (100.00%)	262 (100.00%)	292 (100.00%)
RepairAgent	305	Source (GPT-3.5)	322.90 (12.93%)	117 (63.24%)	88 (63.31%)	73 (63.48%)
		Hotswap	1616.94 (64.73%)	146 (78.92%)	108 (77.70%)	89 (77.39%)
		Target (GPT-4o)	2497.82 (100.00%)	185 (100.00%)	139 (100.00%)	115 (100.00%)

Cost-effectiveness of parallel hotswapping

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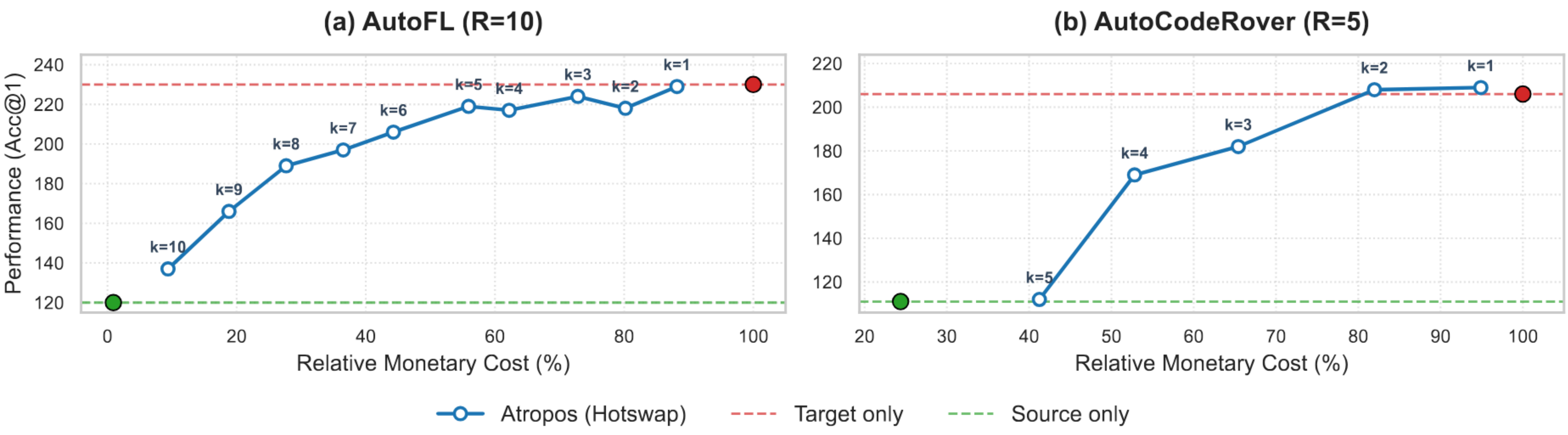

Cost-benefit trade-off of sequential hotswapping

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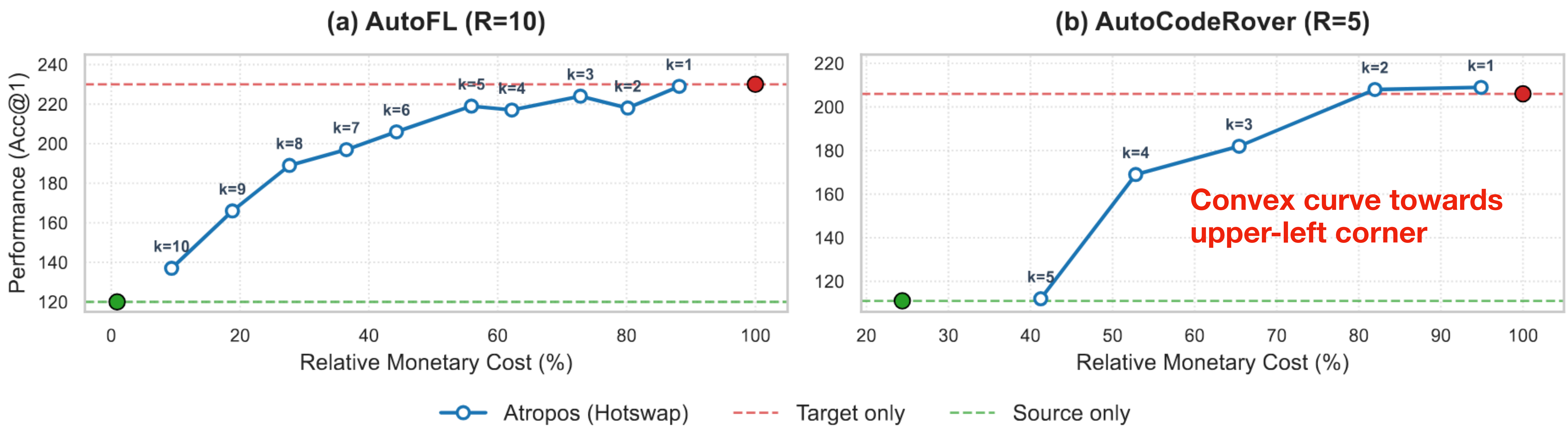

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Cost-benefit trade-off of sequential hotswapping

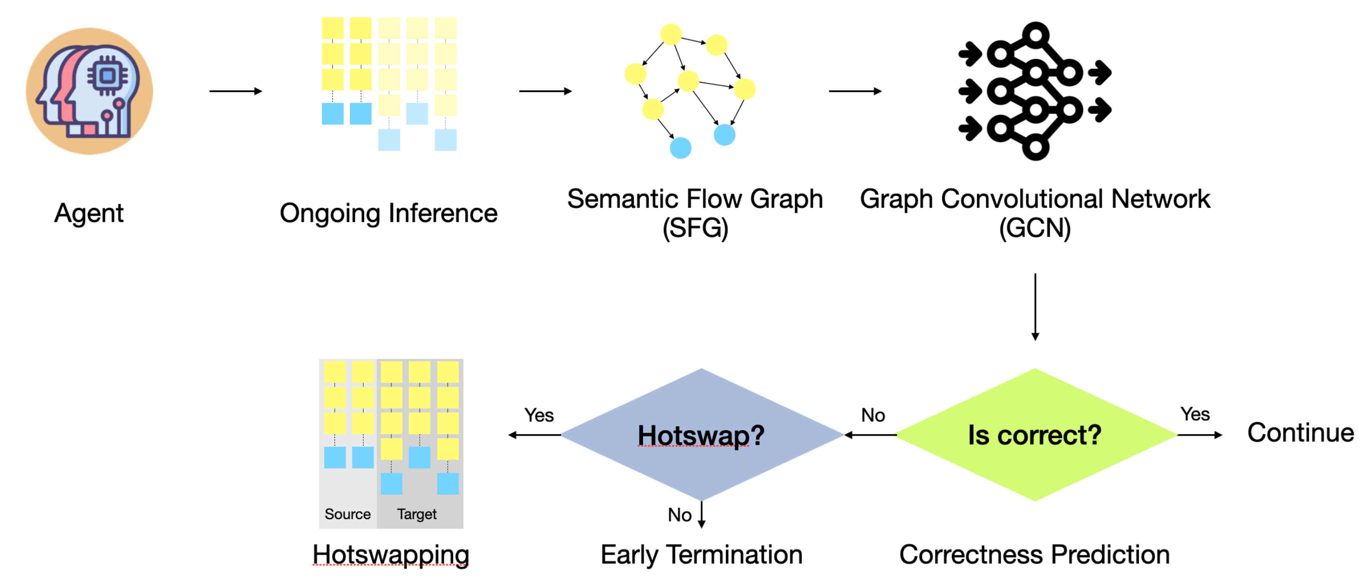
Conclusion

Full paper

Atropos: Improving Cost-Benefit Trade-off of Agents with Early Termination and Model Hotswap

Atropos

A predictive early-termination and LLM hot swapping technique running on a local SLM



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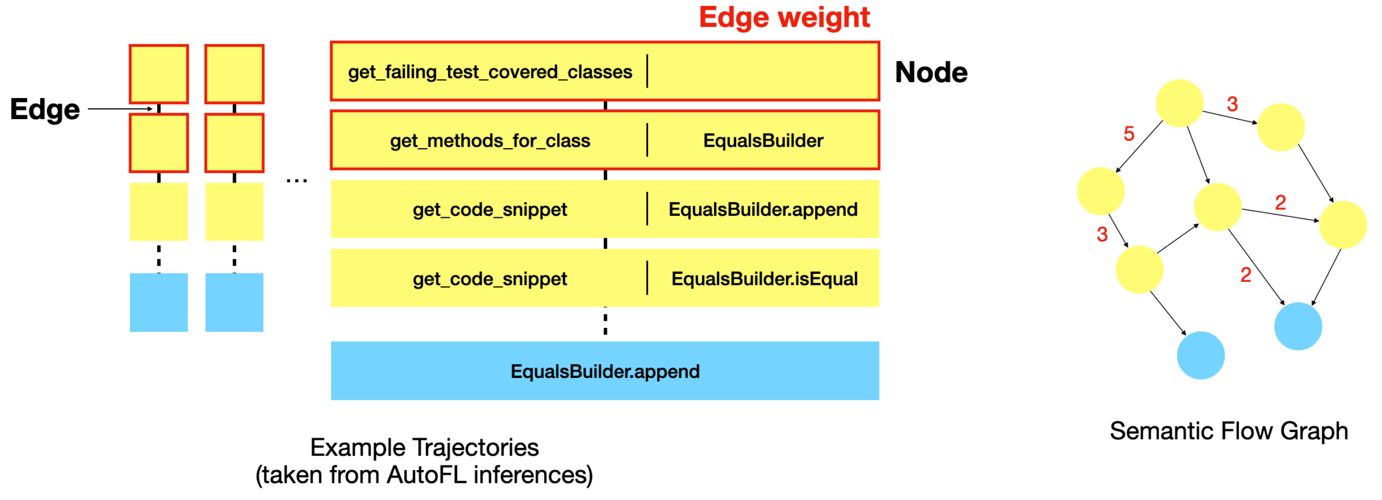

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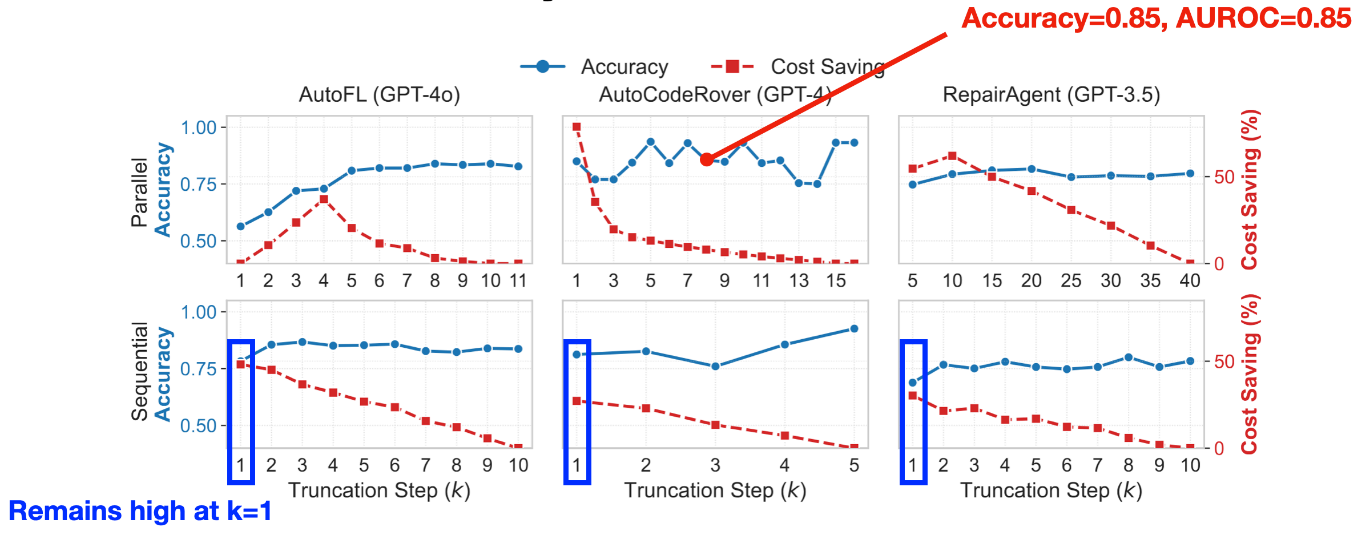

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